SYLLABUS

M.Tech.

Computer Science and Engineering

Batch 2026 onwards



SCHOOL OF COMPUTING

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY UNA

HIMACHAL PRADESH

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Institute – Vision and Mission

Institute Vision:

To build a vibrant multicultural learning environment with value based academic principles and to achieve excellence in teaching and research and to contribute effectively and responsibly to the national and global community.

Institute Mission:

M1: To provide quality education to the students with practical orientation.

M2: To collaborate with industries and research institutions to solve socially relevant problems.

M3: To inspire students to become responsible citizens and competent professionals with ethical values.

School of Computing – Vision and Mission

School of Computing Vision:

To become a center of excellence in emerging areas and train students to become professionals solving challenging societal problems.

School of Computing Mission:

M1: To impart state-of-the-art knowledge in computer science and information technology with emphasis on practical knowledge and to promote specialization in emerging streams.

M2: To participate in research and development in industries and research organizations.

M3: To make the students aware of ethics and apply them in solving social problems.

M.Tech. in Computer Science and Engineering

Program Educational Objectives (PEOs):

PEO1 Career: Pursue a successful career in academics, entrepreneurship, design and development, or research in the domain of Computer Science.

PEO2 Professionalism: Analyze, design and formulate computational research problems by applying in-depth domain knowledge and research principles.

PEO3 Responsibility: Work effectively in multi-disciplinary and multi-cultural environments by respecting professionalism, ethical and social norms; along with environmental awareness at national and international level.

PEO3 Life-long learning: Engage in lifelong learning to always be an active learner towards the technical professional growth long with betterment of the society and country.

PEO to Mission Mapping:

Mission / PEO	M1	M2	M3
PEO1	3	3	1
PEO2	3	2	3
PEO3	1	2	3
PEO4	2	2	3

Program Outcomes (POs):

PO1: An ability to independently carry out research / investigation and development work to solve practical problems.

PO2: An ability to write and present a substantial technical report/ document & research outcome.

PO3: Students should be able to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a higher level than the requirements in the appropriate bachelor program.

Program Specific Outcomes (PSOs):

PSO1: Expertise in advanced computer science domains such as Artificial Intelligence, Data Science, or Cybersecurity.

PSO2: To conduct high-quality research and contribute to the advancement of knowledge in specialized areas of computer science such as data science or cybersecurity.

PSO3: To design and develop scalable, efficient, and secured computing models for real-life applications.

PO/PSO to PEO Mapping:

	PEO1	**PEO2**	**PEO3**	**PEO4**
PO1	2	3	2	3
PO2	1	3	2	1
PO3	1	3	2	3
PSO1	3	2	1	1
PSO2	2	3	2	3
PSO3	1	2	3	3

Choice of Specialization

The curriculum offers two specializations, viz., Data Science and Cyber Security. The students have to opt for any one specialization as per their interest in the first semester itself and the course registration has to be done accordingly. The specialization once chosen will remain the same throughout the program of study and in between it cannot be changed.

Coding Scheme

Example

M	C	P	C	5	0	1
1	2	3	4	5	6	7

The 7 letter code represents the following:	
Letter 1:	Represents the Master degree program.
Letter 2:	Denotes one of the following: 'C' for CSE Branch courses, 'I' for Institute level courses, 'D' for Data Science specialized courses, and 'Y' for Cyber Security specialized courses.
Letter 3 & 4:	Curriculum Component such as PC (Program Core), PE (Program Elective), OE (Open Elective), VA (Value Added), and PR (Project / Dissertation).
Letter 5:	Year (level) of the course 5 and 6
Letter 6 & 7:	Course identification number such as 01, 02, etc.

Curriculum Components

S.No.	Category	Credits	Credits(%)
1.	Program Core (PC)	20	34
2.	Program Elective (PE)	12	20
3.	Open Elective (OE)	03	05
4.	Value Added (VA)	02	03
5.	Project and Dissertation (PR)	23	38
Total		60	100

M. Tech. (CSE) with specialization in Data Science/Cyber Security

Program Core (PC) Courses (20 credits)

S.No.	Course Code	Course Name	Credits
1.	MCPC501	Mathematical Foundations for Computing	04
2.	MCPC502	Applied Machine Learning	04
3.	MCPC503	Advanced Algorithms	04
4.	MCPC504	Computability and Complexity	04
5.	MCPC505	Distributed Systems	04
Total			20

Program Elective (PE) Courses (12 credits)[#]

S.No.	Course Code	Course Name	Credits
1.	MXPE5XX	Program Elective I	03
2.	MXPE5XX	Program Elective II	03
3.	MXPE5XX	Program Elective III	03
4.	MXPE5XX	Program Elective IV	03
Total			12

Open Elective (OE) Courses (3 credits)[#]

S.No.	Course Code	Course Name	Credits
1.	MCOE5XX	Open Elective	03
Total			03

[#]A faculty member may also offer a self-designed course as a program/open elective subject to the approval of the competent authority or may be offered through NPTEL/SWAYAM/MOOCs etc.

Value Added (VA) Courses (2 credits)

S.No.	Course Code	Course Name	Credits
1.	MIVA501	Research Methodology	02
2.	MIVA502	Technical Writing and Communication Skills	00
Total			02

Project and Dissertation (PR) (23 credits)

S.No.	Course Code	Course Name	Credits
2.	MCPR501	Minor Project	02
3.	MCPR601	Dissertation I	09
4.	MCPR602	Dissertation II	12
Total			23

SEMESTER-WISE CURRICULUM

L-T-P-C Notation

L-T-P-C $\Rightarrow$ *Lecture-Tutorial-Practical-Credit.*

Credit structure of each course is given in L-T-P-C form (e.g., 2-1-0-3). The numbers corresponding to L, T and P denote the contact hours per week for Lecture, Tutorial and Practical respectively, and that of C denotes the total number of credits for that course in a semester.

SEMESTER I

S.No.	Code	Course Name	L	T	P	C
1.	MCPC501	Mathematical Foundations for Computing	3	1	0	4
2.	MCPC502	Applied Machine Learning	3	0	2	4
3.	MCPC503	Advanced Algorithms	3	1	0	4
4.	MIVA501	Research Methodology	2	0	0	2
5.	MXPE5XX	Program Elective I	3	0	0	3
6.	MXPE5XX	Program Elective II	3	0	0	3
Total			17	2	2	20

SEMESTER II

S.No.	Code	Course Name	L	T	P	C
1.	MCPC504	Computability and Complexity	3	1	0	4
2.	MCPC505	Distributed Systems	3	0	2	4
3.	MXPE5XX	Program Elective III	3	0	0	3
4.	MXPE5XX	Program Elective IV	3	0	0	3
5.	MCOE5XX	Open Elective I$^{\#}$	3	0	0	3
6.	MIVA502	Technical Writing and Communication Skills	1	0	2	0
7.	MCPR501	Minor Project	0	0	4	2
Total			16	1	8	19

SEMESTER III

S.No.	Code	Course Name	L	T	P	C
1.	MCPR601	Dissertation I	-	-	-	9
Total			-	-	-	9

SEMESTER IV

S.No.	Code	Course Name	L	T	P	C
1.	MCPR602	Dissertation II	-	-	-	12
Total			-	-	-	12

#Courses may be offered through NPTEL/SWAYAM/MOOCs etc.

Summary					
Semester	I	II	III	IV	Total
Credit	20	19	9	12	60

Semester-wise Program Electives
(Specialization in Data Science)

Program Elective I (Semester I)		
S.No.	**Code**	**Course Name**
1.	MDPE501	Introduction to Data Analytics
2.	MDPE502	Advanced Statistics
3.	MDPE503	Soft Computing

Program Elective II (Semester I)		
S.No.	**Code**	**Course Name**
1.	MDPE504	Data Mining
2.	MDPE505	Digital Image and Multimedia Processing
3.	MDPE506	Principles of Internet of Things

Program Elective III (Semester II)		
S.No.	**Code**	**Course Name**
1.	MDPE507	Big Data Analytics
2.	MDPE508	Computer Vision Algorithms
3.	MDPE509	Natural Language Processing

Program Elective IV (Semester II)		
S.No.	**Code**	**Course Name**
1.	MDPE510	Advanced Deep Networks
2.	MDPE511	Applied Accelerated Artificial Intelligence
3.	MDPE512	Large Language Models

Semester-wise Program Electives
(Specialization in Cyber Security)

Program Elective I (Semester I)		
S.No.	**Code**	**Course Name**
1.	MYPE501	Information Theory and Coding
2.	MYPE502	Network and Database Security
3.	MYPE503	Modern Cryptography

Program Elective II (Semester I)		
S.No.	**Code**	**Course Name**
1.	MYPE504	Penetration Testing and Vulnerability Assessment
2.	MYPE505	Next-Generation Computer Networks
3.	MYPE506	Secure Computation

Program Elective III (Semester II)		
S.No.	**Code**	**Course Name**
1.	MYPE507	Malware Analysis and Intrusion Detection
2.	MYPE508	Mobile Security
3.	MYPE509	Blockchain Technologies

Program Elective IV (Semester II)		
S.No.	**Code**	**Course Name**
1.	MYPE510	Ethical Hacking
2.	MYPE511	Digital Forensics
3.	MYPE512	Responsible AI

SEMESTER I

Course Code	MCPC501
Course Title	Mathematical Foundations for Computing
L-T-P-C	3-1-0-4
Course Type	PC

COURSE OBJECTIVES

- Understand fundamental concepts in logic, set theory, relations, and functions to formulate precise mathematical arguments.
- Develop algebraic reasoning through groups, rings, vector spaces, and linear transformations.
- Grasp probability theory and apply it to model and analyze random processes.
- Analyze structures and algorithms related to graph theory and their computational implications.
- Apply number theory concepts to cryptography and computing systems.

COURSE CONTENT

Unit-I: Logic and Set Theory
Proposition Logic, Resolution Proof system, Predicate logic; Sets, subsets, operations, Venn diagrams; Relations: Properties, equivalence relations, partial orderings; Functions: Injective, surjective, bijective, inverse, compositions.

Unit-II: Algebra
Groups, homomorphism theorems, cosets and normal subgroups, Lagrange's theorem, Ring, Field. Linear algebra: Vector Space, Basis, Matrices and Linear Transformations, Eigen values, Orthogonality.

Unit-III: Probability and Random Variables
Counting, Probability, Discrete random variable, Continuous random variable, Moment generating function, Markov's inequality, Chebyshev's inequality, Geometric distribution, Binomial distributions, Normal Distribution and Central Limit Theorem.

Unit-IV: Graph Theory
Graphs, Euler tours, planar graphs, Hamiltonian graphs, Euler's formula, applications of Kuratowski's theorem, graph colouring, chromatic polynomials, trees, weighted trees, the max-flow min-cut theorem.

Unit-V: Number Theory
Divisibility, primes, Euclidean algorithm, Modular arithmetic, Congruences, Fermat's theorem, Euler function, Chinese remainder theorem, Applications in cryptography.

COURSE OUTCOMES

CO	Description	Level
CO1	Apply formal logic, set theory, and function properties to solve computing problems	K3
CO2	Demonstrate understanding of algebraic structures and linear algebra for theoretical computation models	K2
CO3	Analyze probabilistic models and distributions relevant to computing scenarios	K4
CO4	Explain fundamental concepts in graph theory and apply them in computing applications	K2
CO5	Solve number-theoretic problems using modular arithmetic and apply to cryptographic protocols	K2

TEXT BOOKS

1. Donald F. Stanat and David F. McAllister, *Discrete Mathematics in Computer Science*, 1977.
2. Thomas Koshy, *Elementary Number Theory with Applications*, Elsevier, 2002.
3. I.N. Herstein, *Topics in Algebra*, John Wiley & Sons, 1990.
4. Sheldon M. Ross, *Introduction to Probability Models*, Elsevier, 2014.
5. G. Chartrand and P. Zhang, *Introduction to Graph Theory*, McGraw-Hill Companies, 2006.
6. Kenneth Hoffman and Ray Kunze, *Linear Algebra*, 2nd Edition, PHI Learning, 2009.

REFERENCE BOOKS

1. Douglas B. West, *Introduction to Graph Theory*, Prentice Hall of India, 2001.
2. Niven and Zuckerman, *An Introduction to the Theory of Numbers*,1972.
3. Castaneda, L. B., Arunachalam, V., Dharmaraja, D., *Introduction to Probability and Stochastic Processes with Applications*, John Wiley & Sons, Inc, Publications, 2012.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PSO1	PSO2	PSO3
CO1	3	3	3	3	-	-
CO2	2	3	3	3	2	-
CO3	3	2	-	-	2	-
CO4	-	2	-	-	2	-
CO5	3	2	3	3	2	-
Score	11	12	9	9	8	-
COM	3	2	3	3	2	-

Course Code	MCPC502
Course Title	Applied Machine learning
L-T-P-C	3-0-2-4
Course Type	PC

COURSE OBJECTIVES

- To develop a foundational understanding of machine learning paradigms and learning from data using statistical and computational methods.
- To equip students with the ability to model, analyze, and evaluate machine learning algorithms for both supervised and unsupervised tasks
- To impart knowledge about neural networks, support vector machines, decision trees, and fuzzy inference systems as intelligent learning models.
- To introduce clustering techniques and data transformation methods essential for high-dimensional data analysis and reduction.
- To familiarize students with real-world applications of machine learning in business intelligence, big data analytics, and data mining.

COURSE CONTENT

Unit-I: Introduction
Towards Intelligent Machines, Machine Learning Problems, Examples and Applications, Data Representation, Time Series Forecasting, Domain Knowledge, Diversity of Data, Forms of Learning, Supervised Learning: Learning from Observations, Bias and Variance, Why Learning Works, Overfitting Avoidance, Inductive Learning, Estimating Generalization Errors, Metrics for Assessing Regression and classification.

Unit-II: Statistical Learning and Support Vector Machines
Statistical Learning : Machine Learning and Inferential Statistical Analysis, Descriptive Statistics, Bayesian Reasoning, k-Nearest Neighbor (k-NN) Classifier, Discriminant Functions and Regression Functions, Linear Regression with Least Square Error Criterion, Logistic Regression for Classification Tasks, Support Vector Machines (SVM): Introduction, Linear Discriminant Functions for Binary Classification, Perceptron Algorithm, Linear Maximal Margin, Linear Soft Margin Classifier, Kernel-Induced Feature Spaces, Nonlinear Classifier, Regression using SVM, Decomposing Multiclass Classification Problem Into Binary Classification Tasks.

Unit-III: Learning With Neural Networks and Decision Tree
Neural Networks: Cognitive Machine, Neuron Models, Network Architectures, Perceptron, Linear Neuron and Learning Rule, Multi-Layer Perceptron (MLP) Networks, Error-Backpropagation Algorithm, Multi-Class Discrimination with MLP Networks, Radial Basis Functions

(RBF) Networks, Decision Tree Learning: Introduction, Measures of Impurity for Evaluating Splits in Decision Trees, ID3, C4.5, and CART Decision Trees, Pruning the Tree, Strengths and Weaknesses of Decision-Tree Approach.

Unit-IV: Data Clustering and Data Transformations
Unsupervised Learning, Engineering the Data, Overview of Basic Clustering Methods, K-Means Clustering, Fuzzy K-Means Clustering, Expectation-Maximization (EM) Algorithm and Gaussian Mixtures Clustering, Data Transformations , Entropy-Based Method for Attribute Discretization, Principal Components Analysis (PCA) for Attribute Reduction, Rough Sets-Based Methods for Attribute Reduction.

Unit-V: Business Intelligence and Data Mining
Introduction to Analytics , CRISP-DM Model, Data Warehousing and Online Analytical Processing, Mining Frequent Patterns and Association Rules, Intelligent Information Retrieval Systems, Applications and Trends, Technologies for Big Data, Fuzzy Inference Systems , Genetic Algorithm (GA) for Search Optimization.

LAB CONTENT

The experiments may be performed as per the course contents mentioned above.

COURSE OUTCOMES

CO	Description	Level
CO1	Understand key machine learning concepts, forms of learning, and evaluation metrics for regression and classification tasks	K2
CO2	Apply statistical learning methods, including k-NN, regression, Bayesian reasoning, and support vector machines for predictive modeling	K2
CO3	Design and implement neural network models and decision tree algorithms for classification and pattern recognition	K4
CO4	Perform data clustering and dimensionality reduction using unsupervised learning and transformation techniques like PCA and EM	K3
CO5	Demonstrate applications of machine learning in business intelligence, data mining, and optimization using GA and fuzzy inference systems	K4

TEXT BOOKS

1. Gopal, Madan., *Applied machine learning*, McGraw-Hill Education, 2019.
2. Tom, Mitchell, *Machine Learning*, McGraw-Hill, 2017.

REFERENCE BOOKS

1. Ethem, Alpaydin, *Introduction to Machine Learning*, PHI, 2005.
2. Géron, Aurélien., *Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow.*, O'Reilly Media, Inc., 2022.
3. Goodfellow, Ian, et al, *Deep learning. Vol. 1. No. 2.*, Cambridge: MIT press, 2016.

CO to PO/PSO Mapping

	PO1	**PO2**	**PO3**	**PSO1**	**PSO2**	**PSO3**
CO1	3	2	3	3	3	3
CO2	3	2	3	3	3	3
CO3	3	2	3	3	3	3
CO4	1	3	3	2	3	1
CO5	3	2	3	3	3	3
Score	13	11	15	14	15	13
COM	3	2	3	3	3	3

Course Code	MCPC503
Course Title	Advanced Algorithms
L-T-P-C	3-1-0-4
Course Type	PC

COURSE OBJECTIVES

- To analyze the asymptotic performance of algorithms.
- To demonstrate a familiarity with major algorithms and data structures.
- To apply important algorithmic design paradigms and methods of analysis.
- To understand the fundamentals of approximation and randomized algorithms.
- To synthesize efficient algorithms in common engineering design applications.

COURSE CONTENT

Unit-I: Introduction
Efficiency analysis and Asymptotic analysis. Complexity of algorithms and Algorithmic paradigms, Sorting and Searching techniques with analysis, Greedy algorithms, divide and conquer, dynamic programming, applications

Unit-II: Graph Algorithms
Definition, terminology, directed and undirected graphs, properties, implementation of adjacency matrix and linked adjacency chains, connectivity in graphs, graph traversal, breadth first and depth first, spanning trees, Graph algorithms: DFS and BFS with applications, MST, shortest paths, matchings

Unit-III: Dynamic Programming and Maximum Flows
Longest Increasing Subsequence, Edit Distance, Subset Sum, Optimal BSTs, Flows, Cuts, Maxflow-Mincut, Augmenting Paths, Bipartite Matchings, Other Settings, Applications of Flows: Exam Scheduling, Baseball Elimination, Project Selection.

Unit-IV: Approximation and Randomized Algorithms
Introduction to Approximation Frameworks, Vertex Cover via Maximal Matchings, Vertex Cover via LP rounding, TSP, Set Cover, Monte Carlo v. Las Vegas, Min-Cut Algorithm, MAX SAT via the Probabilistic Methods, 2SAT via Markov Chains, Primality Testing.

Unit-V: Parameterized and Exact Algorithms
Closest String, Iterative Compression for FVS, Randomized Algorithm for k-Path, DP over subsets - Set Cover, Branch and Bound, An Inclusion-Exclusion approach to Hamiltonian Path, Dynamic Programming for TSP, Local Search.

COURSE OUTCOMES

CO	Description	Level
CO1	Understand and analyze the performance of the fundamentals of algorithms.	K3
CO2	Illustrate the applications of graph algorithms in real world.	K3
CO3	Explain the dynamic programming and maximum flow.	K4
CO4	Analyze the performance of randomized and approximation algorithm.	K3
CO5	Demonstrate a familiarity with applied algorithmic settings.	K3

TEXT BOOKS

1. Cormen, Leiserson, Rivest, and Stein, *Introduction to Algorithms*. 2nd ed. Cambridge, MA: MIT Press, 2001..
2. Williamson, David P., and David B. Shmoys., *The Design of Approximation Algorithms*. Cambridge University Press, 2011.

REFERENCE BOOKS

1. Dasgupta, S., Papadimitriou, C. H., and Vazirani, U. V. (2008), *Algorithms*, McGraw-Hill Higher Education, 2008.
2. Kruse, R., and Tondo, C. L., *Data structures and program design in C*, Pearson Education India, 2007.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PSO1	PSO2	PSO3
CO1	2	2	-	3	-	3
CO2	-	-	-	3	-	3
CO3	2	2	-	-	-	-
CO4	2	2	-	-	-	-
CO5	2	2	-	-	-	3
Score	8	8	-	6	-	9
COM	2	2	-	3	-	3

Course Code	MIVA501
Course Title	Research Methodology
L-T-P-C	2-0-0-2
Course Type	VA

COURSE OBJECTIVES

- To understand the fundamentals of research, its objectives, types, and methodology.
- To understand the significance of literature review, and identifying gaps.
- To learn various statistical techniques for data analysis, interpretation, tabulation, and graphical representation.
- To learn the principles and formats of effective report writing, structuring, and documenting research findings.
- To understand ethical principles and standards in research, including intellectual property rights, and scholarly publishing practices.

COURSE CONTENT

Unit-I: Foundation of Research
What is Research?, Objective of Research, Scientific Research, Research and Theory, conceptual and Theoretical Models, Importance of Research Methodology in Scientific Research. Classification of Research: Pure, Applied, Experimentals.

Unit-II: Literature Survey and Problem Definition
Importance of literature review in defining a problem, literature review-primary and secondary sources, reviews, monograph, patents, research databases, web as a source, critical literature review, identifying gap areas from literature and research database, development of working hypothesis.

Unit-III: Data Processing and Statistical Analysis
Editing, Classification and Coding, Transcription, Tabulation, and Graphical Representation, Measures of Relationship, Simple Regression Analysis, Multiple Correlation and Regression, Partial Correlation. Statistical Analysis, Measures of Central Tendency, Measures of Dispersion, Measures of Association/Relationship, Probability distributions: Binomial, Poisson, Uniform, Normal and Exponential, Hypothesis Testing, Confidence Interval, Test of Significance, Comparison of Two Proportions, Comparison of Means (z-test, and t-test).

Unit-IV: Report Writing
Types of Reports, Planning of Report Writing, Research Report Format, Principles of Writing, Documentation, Data and Data Analysis Reporting in a Thesis, Writing of Report, Typing of

Report.

Unit-V: Research Ethics, IPR and Scholarly Publishing
Ethics- ethical issues, ethical committees (human and animal); IPR- intellectual property rights and patent law, commercialization, copy right, royalty, trade related aspects of intellectual property rights (TRIPS); scholarly publishing- IMRAD concept and design of research paper, citation and acknowledgment, plagiarism, reproducibility and accountability.

COURSE OUTCOMES

CO	Description	Level
CO1	Understand and apply various research methodologies	K3
CO2	Define and refine a research problem and Identify gaps in existing knowledge.	K2
CO3	Apply statistical techniques to analyze and interpret data, and effectively present findings.	K4
CO4	Write a clear, well-structured, and effective research report.	K2
CO5	Apply ethical principles and standards in research.	K2

TEXT BOOKS

1. R. Barker Bausell, *Discrete Mathematics and Its Applications*, Scarecrow Press, 2013.
2. Wayne Goddard,Stuart Melville, *Research Methodology: An Introduction*, Juta and Co. Ltd., 2007.

REFERENCE BOOKS

1. C.R. Kothari, *Research Methodology: Methods and Techniques*, New Age International Publishers, 2004.
2. Fisher R.A, *Statistical Methods for Research Workers*, Cosmo Publications. Ltd., 2007.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PSO1	PSO2	PSO3
CO1	2	3	2	-	2	-
CO2	2	3	2	-	-	-
CO3	2	2	2	-	-	-
CO4	2	2	2	-	-	-
CO5	2	2	2	-	-	-
Score	10	12	10	-	2	-
COM	2	2	2	-	2	-

Course Code	MDPE501
Course Title	Introduction to Data Analytics
L-T-P-C	3-0-0-3
Course Type	PE

COURSE OBJECTIVES

- To understand and apply basic statistical methods to summarize data and make inferences.
- To be able to perform data cleaning, transformation, and feature engineering for effective analysis.
- To be able to build, interpret, and evaluate regression models for predictive analysis.
- To create clear and effective data visualizations using appropriate tools and techniques.
- To explore advanced trends in analytics, including AI integration and real-time data processing.

COURSE CONTENT

Unit-I: Descriptive and Inferential Statistics
Introduction to statistics: its role and importance, Descriptive statistics: summarizing data (central tendency, dispersion, distribution), Introduction to Probability, Random Variables and Probability Distributions, Inferential Statistics: Motivation, Single sample tests, Two Sample tests, Type 1 and Type 2 Errors, Confidence Intervals, ANOVA and Test of Independence.

Unit-II: Statistical Analysis and Transformation
Data Cleaning, Data Transformation: Normalization, Standardization, Encoding of categorical data, Feature Engineering: Extracting Features and dimensionality reduction using PCA, Data Integration, Data Storage and Retrieval.

Unit-III: Regression Analysis
Correlation: measuring the strength of linear relationships, Simple linear regression: Modelling relationships between variables, Model evaluation: assessing goodness of fit, Multiple Linear Regression, Equation of multiple linear regression, Interpretation of multiple linear regression, F-test for Multiple Linear Regression, t-tests in Multiple Linear Regression, Cautions about Regression.

Unit-IV: Data Visualization and Analysis Techniques
Principles of effective visualization - Types of visualizations: Bar charts, Line graphs, Scatter plots, Heatmaps, Visualization of large-scale and multi-dimensional data, Geographic Information System (GIS) visualizations, Interactive visualizations: Dashboards and dynamic reports,

Data Visualization Tools.

Unit-V: Emerging Topics in Data Analytics
AI and Machine Learning Integration, Explainable AI (XAI), Federated Learning, Real Time Data Analytics, Streaming Analytics, Data Visualization Innovations, Augmented and Virtual Reality in Data Visualization, AI in Data Visualization: AutoML and augmented analytics.

COURSE OUTCOMES

CO	Description	Level
CO1	Apply descriptive and inferential statistical techniques to analyze and interpret data effectively.	K3
CO2	Prepare and transform raw data using cleaning, normalization, encoding, and dimensionality reduction methods.	K3
CO3	Develop and assess linear regression models to explain relationships and make predictions.	K3
CO4	Design and implement impactful data visualizations to communicate insights clearly.	K3
CO5	Demonstrate understanding of emerging data analytics technologies such as AI, real-time analytics, and advanced visualization tools.	K2

TEXT BOOKS

1. Trevor Hastie, Robert Tibshirani, and Jerome Friedman, *The Elements of Statistical Learning*, Vol. 2, No. 1, Springer, New York, 2009.
2. Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani, *An Introduction to Statistical Learning with Applications in R*, Springer, 2021.
3. Colin Ware, *Information Visualization: Perception for Design*, 4th Edition, Elsevier, 2019.

REFERENCE BOOKS

1. Cole Nussbaumer Knaflic, *Storytelling with Data: A Data Visualization Guide for Business Professionals*, Wiley, 2015.
2. Hadley Wickham and Garrett Grolemund, *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*, 2nd Edition, O'Reilly Media, 2023.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PSO1	PSO2	PSO3
CO1	3	2	3	3	3	3
CO2	3	2	3	3	3	3
CO3	3	2	3	3	3	3
CO4	1	3	3	2	3	1
CO5	3	2	3	3	3	3
Score	13	11	15	14	15	13
COM	3	2	3	3	3	3

Course Code	MDPE502
Course Title	Advanced Statistics
L-T-P-C	3-0-0-3
Course Type	PE

COURSE OBJECTIVES

- To study fundamentals of statistics concepts, including data types and its various study design used in research.
- To explore various data analytics suing visualization technique and apply appropriate inference methods for hypothesis testing and estimation.
- To explore various regression models and ANOVA techniques to analyze relationship among variables or groups.
- To explore various techniques to calculate probability and survival
- To study various statistical learning methods to obtain valid conclusions and to highlight common statistical pitfalls and ethical considerations.

COURSE CONTENT

Unit-I: Foundation of Statistics
Data types: binary, continuous, categorical, ordinal, count, survival, longitudinal with examples. Study types: prospective, retrospective, case-control, cohort, cross-sectional, qualitative studies, randomized controlled trials.

Unit-II: Data analysis and Statistical inference
Exploratory Data Analysis: data visualization and statistical plots. Statistical Inference: Estimation, Hypothesis testing. Useful Statistical Tests: z-test, t-test, F-test, Chi-Square test, Wilcoxon signed-rank test with applications.

Unit-III: Applied regression and Analysis of Variance
Regression: linear regression, logistic regression, count regression. Analysis of Variance (ANOVA): One-way and two-way ANOVA.

Unit-IV: Statistical probability
Bayesian Techniques: Bayes theorem, prior and posterior distributions, Bayesian inference. Survival Analysis: survival function, hazard function, Kaplan–Meier curves, log-rank test, Cox proportional hazards regression.

Unit-V: Statistical learning and ethics
Statistical Learning: clustering and classification. Statistical traps to avoid: correlation vs causation, misuse of p-values and multiple testing, handling outliers, misuse of variable selection,

misuse of data visualization, ethical issues in statistics. Case studies with real data.

COURSE OUTCOMES

CO	Description	Level
CO1	Identify and differentiate between various data types and classify different study types based on the situation.	K2
CO2	Conduct exploratory data analysis using appropriate data visualization and statistical plots, and apply relevant statistical tests to perform hypothesis testing and draw inferences from data.	K3
CO3	Implement and interpret different regression models and perform one-way and two-way ANOVA for data analysis.	K4
CO4	Apply Bayes' theorem to derive posterior distributions and conduct basic Bayesian inference, as well as calculate and interpret survival and hazard functions.	K4
CO5	Apply statistical learning technique, and identify common statistical traps in statistical analysis.	K5

TEXT BOOKS

1. Wasserman, L. All of statistics: a concise course in statistical inference. Vol. 26. New York: Springer, 2004.
2. Hogg, R.V., Tanis, E.A. and Zimmerman, D.L. Probability and Statistical Inference, 192. Printice Hall, Upper Saddle River, NJ (2014).
3. Gareth, J., Daniela, W., Trevor, H., and Robert, T. An introduction to statistical learning: with applications in R. Spinger, 2013
4. Weisberg, S. Applied linear regression. Vol. 528. John Wiley & Sons, 2005
5. Kleinbaum, D.G., and Klein M. Survival analysis a self-learning text. Springer, 2011

REFERENCE BOOKS

1. Koch, K.R. Introduction to Bayesian statistics. Springer Science & Business Media, 2007
2. Motulsky, H. Intuitive Biostatistics: a Nonmathematical guide to Statistical Thinking. Oxford University Press, USA, 2014

3. Kabacoff, R. Data visualization with R. Quantitative Analysis Center: Wesleyan University, 2020

4. Montgomery, D.C., Peck, E.A., and Vining, G.G. Introduction to Linear Regression Analysis, 5th Ed., Wiley, 2012

5. Rohatgi, V.K. and Saleh, A.K. An Introduction to Probability and Statistics, 3rd Ed., Wiley, 2015.

CO to PO/PSO Mapping

	PO1	**PO2**	**PO3**	**PSO1**	**PSO2**	**PSO3**
CO1	2	1	3	3	1	1
CO2	3	2	3	3	2	2
CO3	3	2	3	3	2	3
CO4	3	2	3	3	3	2
CO5	3	2	3	3	3	3
Score	14	9	15	15	11	11
COM	3	2	3	3	2	2

Course Code	MDPE503
Course Title	Soft Computing
L-T-P-C	3-0-0-3
Course Type	PE

COURSE OBJECTIVES

- Understand the theoretical foundation and differences between soft and hard computing techniques.
- Explore and implement various neural network architectures and learning paradigms.
- Analyze and design fuzzy inference systems and fuzzy control mechanisms for industrial applications.
- Apply evolutionary and bio-inspired algorithms to solve optimization problems.
- Integrate hybrid intelligent systems and develop real-time control solutions for manufacturing and distributed systems.

COURSE CONTENT

Unit-I: Foundations of Soft Computing and Neural Networks
Introduction to Soft Computing: Hard vs Soft Computing, Adaptive systems and update mechanisms, Applications in engineering, manufacturing, and management, Artificial Neural Networks (ANN): Basic concepts, biological analogy, Multi-layer feedforward networks, Backpropagation algorithm, Recurrent Neural Networks (RNN), Reservoir Computing, Learning paradigms: supervised, unsupervised, reinforcement learning, Functional Link ANN (FLANN), Radial Basis Function (RBF) networks.

Unit-II: Fuzzy Logic and Fuzzy Control Systems
Classical sets vs. fuzzy sets, fuzzy relations, Membership Functions: properties and optimization, Fuzzy inference systems: Mamdani & Takagi-Sugeno (TS) models, Fuzzification and defuzzification methods, Fuzzy control from an industrial perspective, Knowledge-based systems for process control, Non-linear and adaptive fuzzy control systems, Applications in manufacturing, embedded systems, process automation.

Unit-III: Genetic Algorithms and Evolutionary Computing
Genetic Algorithms (GA): Introduction, mathematical foundation, Encoding, selection, crossover, mutation, elitism, Genetic-based machine learning, Comparison with conventional optimization techniques, Differential Evolution (DE), Particle Swarm Optimization (PSO), Ant Colony Optimization (ACO), Bacterial Foraging Algorithm (BFA), Multi-objective optimization: NSGA-II, MOPSO, Applications: scheduling, layout design, clustering, design optimization.

Unit-IV: Chaos Theory and Self-Organizing Systems
Basics of chaos and complexity: order vs disorder, Evolution of possibilities and strange attractors, Deterministic chaos: Lorenz system, logistic map, Chaos in real systems: manufacturing, energy, and control systems, Self-organization, emergence, and synergetics, Modeling and analyzing chaotic behavior in neural and fuzzy systems.

Unit-V: Evolutionary Dynamics, Hybrid Systems and Real-Time Applications
Evolutionary dynamics in problem solving, Hybrid intelligent systems: Neuro-Fuzzy, GA-Fuzzy, ANN-GA, Evolutionary engineering: optimization in CAD/CAM, manufacturing, Real-time distributed systems and control: Software development using CORBA, Java, XML, Agent-based collaboration and intelligent systems, Neural network and fuzzy logic for process control, Integration with PLCs, SCADA, and IoT systems, Case studies in production optimization, fault detection, and smart automation.

COURSE OUTCOMES

CO	Description	Level
CO1	Understand core principles of neural networks, fuzzy systems, and evolutionary algorithms	K2
CO2	Apply soft computing models to nonlinear and real-time control systems	K3
CO3	Analyze and simulate chaotic and self-organizing systems in manufacturing and control	K4
CO4	Analyze hybrid intelligent systems using bio-inspired and fuzzy techniques	K4
CO5	Analyze the application of emerging technologies in distributed and intelligent control systems	K4

TEXT BOOKS

1. Principles of Soft Computing by S.N. Sivanandam and S.N. Deepa, Wiley India, 2018.
2. Neuro-Fuzzy and Soft Computing by J.-S. R. Jang, C.-T. Sun, and E. Mizutani, Pearson, 2015.

REFERENCE BOOKS

1. Multi-Objective Optimization using Evolutionary Algorithms by Kalyanmoy Deb, Wiley, 2001.
2. Swarm Intelligence by James Kennedy and Russell Eberhart, Morgan Kaufmann, 2001.

3. Neural Networks and Learning Machines by Simon Haykin, Pearson, 2009.
4. Chaos and Nonlinear Dynamics by Robert C. Hilborn / H.T. Banks et al, 2001.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PSO1	PSO2	PSO3
CO1	2	2	3	2	1	1
CO2	2	2	3	2	1	2
CO3	2	1	2	1	2	1
CO4	2	2	3	2	2	2
CO5	1	2	2	1	2	2
Score	9	9	13	8	8	8
COM	2	2	3	2	2	2

Course Code	MDPE504
Course Title	Data Mining
L-T-P-C	3-0-0-3
Course Type	PE

COURSE OBJECTIVES

- To understand fundamental concepts and algorithms used in data mining.
- To explore preprocessing, data cleaning, and transformation techniques.
- To analyze methods for classification, clustering, association, and anomaly detection.
- To apply data mining models and algorithms using real-world datasets.
- To evaluate and interpret data mining results for decision-making and knowledge discovery.

COURSE CONTENT

Unit-I: Introduction and Data Preprocessing
Introduction To Data Mining, KDD Process, Types Of Data, Data Quality, Preprocessing Techniques, Data Integration, Data Reduction, Discretization and Concept Hierarchy Generation.

Unit-II: Association Rule Mining
Frequent Itemset Generation, Apriori Algorithm, FP-Growth Algorithm, Interestingness Measures, Advanced Association Rules, Constraint-Based Mining, Mining Multilevel and Multidimensional Patterns.

Unit-III: Classification
Supervised Learning Concepts, Decision Trees, Rule-Based Classifiers, Naïve Bayes, Support Vector Machines, K-Nearest Neighbors, Model Evaluation And Selection, Ensemble Methods.

Unit-IV: Clustering And Anomaly Detection
Types Of Clustering, Partitioning Methods, Hierarchical Clustering, Density-Based Methods (DBSCAN), Grid-Based Methods, Evaluation of Clustering, Distance Measures, Outlier and Anomaly Detection Techniques.

Unit-V: Advanced Topics And Applications
Mining Complex Data Types, Text and Web Mining, Stream Data Mining, Privacy and Security in Data Mining, Applications in Business, Bioinformatics, Cybersecurity, Emerging Trends in Data Mining Research.

COURSE OUTCOMES

CO	Description	Level
CO1	Understand the core concepts and challenges in data mining.	K2
CO2	Apply data preprocessing and transformation techniques to prepare raw data.	K3
CO3	Implement classification and association algorithms for data analysis.	K3
CO4	Analyze clustering and outlier detection results using evaluation metrics.	K4
CO5	Develop and apply data mining models for real-world applications.	K3

TEXT BOOKS

1. Charu C. Aggarwal, Data Mining: The Textbook, 2nd Edition, Springer, 2023
2. Jiawei Han, Micheline Kamber, and Jian Pei, Data Mining: Concepts and Techniques, 4th Edition, Morgan Kaufmann, 2022

REFERENCE BOOKS

1. Pang-Ning Tan, Michael Steinbach, Anuj Karpatne, and Vipin Kumar, Introduction to Data Mining, 2nd Edition, Pearson, 2018
2. Ian H. Witten, Eibe Frank, Mark A. Hall, Data Mining: Practical Machine Learning Tools and Techniques, 4th Edition, Elsevier, 2016

CO to PO/PSO Mapping

	PO1	PO2	PO3	PSO1	PSO2	PSO3
CO1	3	2	2	3	-	-
CO2	2	3	2	2	3	-
CO3	3	2	3	3	2	1
CO4	2	3	2	2	2	2
CO5	3	3	3	3	3	2
Score	13	13	12	13	10	5
COM	3	3	2	3	3	2

Course Code	MDPE505
Course Title	Digital Image and Multimedia Processing
L-T-P-C	3-0-0-3
Course Type	PE

COURSE OBJECTIVES

- To understand fundamental concepts of classical image and multimedia processing with modern data science tools.
- To explore preprocessing, data cleaning, and transformation techniques.
- To analyze methods for classification, clustering, association, and anomaly detection.
- To apply data mining models and algorithms using real-world datasets.
- To evaluate and interpret data mining results for decision-making and knowledge discovery.

COURSE CONTENT

Unit-I: Introduction to Digital Image and Multimedia
Basics of signals: 1D (audio), 2D (image), and 3D (video) signals; multimedia types and formats — image, audio, video — characteristics and differences. Digital representation of multimedia: pixels (image), waveforms (audio), frame sequences (video). Color models for images: RGB, HSV, YUV. Audio properties: sampling rate, frequency, amplitude. Video properties: frame rate, resolution, codecs. Real-world applications in healthcare, surveillance, media, and social platforms. Overview of multimedia datasets: COCO (image), ImageNet (image), YouTube-8M (video), LAION (multimodal).

Unit-II: Multimedia Systems
Architecture of multimedia systems: hardware and software components, Multimedia system workflow: acquisition, processing, storage, retrieval, and presentation, Multimedia file formats and containers: MP4, MP3, AVI, MKV, MOV; Standards and protocols: MPEG, H.264, AAC, RTP , Multimedia data storage and retrieval systems

Unit-III: Multimedia Processing-I
Electromagnetic spectrum and imaging modalities, Classification of images: reflection, emission, and absorption images, Core Operations- Spatial domain techniques, Filtering, Morphological operations, Edge Detection, Color image processing and color space transformations, Compression, Image Restoration

Unit-IV: Multimedia Processing-II
Feature Extraction and Dimensionality Reduction. Audio preprocessing techniques-Trimming

and normalization, Silence removal, Noise reduction, Frame-level operations: frame extraction, resizing, frame differencing

Unit-V: Advanced Topics and Applications
Transfer learning and fine-tuning, Vision-language models: CLIP, BLIP, DALL·E, Flamingo, Prompting and multimodal inference tasks, Applications in Business, Bioinformatics, Cybersecurity, Emerging Trends in Data Mining Research.

COURSE OUTCOMES

CO	Description	Level
CO1	Understand and differentiate between various multimedia types (image, audio, video).	K2
CO2	Explain how multimedia data is digitally represented and processed using pixels, waveforms, and frames.	K2
CO3	Analyze the properties of image, audio, and video content using real-world datasets.	K3
CO4	Preprocess audio and video data using modern data science tools	K4
CO5	Develop and apply visual models for real-world applications.	K4

TEXT BOOKS

1. Rafael C. Gonzalez, Richard E. Woods, Steven L. Eddins, ”Digital Image Processing Using MATLAB,” 2nd ed., Gatesmark Publishing, 2009
2. Li ZN, Drew MS, Liu J. Fundamentals of multimedia. Upper Saddle River (NJ):: Pearson Prentice Hall; 2004.
3. Burger W, Burge MJ. Digital image processing: An algorithmic introduction. Springer Nature; 2022

REFERENCE BOOKS

1. Guan, Ling, ed. Multimedia image and video processing. CRC press, 2017.
2. Parekh R. Principles of multimedia. CRC Press; 2025

CO to PO/PSO Mapping

	PO1	PO2	PO3	PSO1	PSO2	PSO3
CO1	3	2	2	3	-	-
CO2	2	3	2	2	3	-
CO3	3	2	3	3	2	1
CO4	2	3	2	2	2	2
CO5	3	3	3	3	3	2
Score	13	13	12	13	10	5
COM	3	3	2	3	3	2

Course Code	MDPE506
Course Title	Principles of Internet of Things
L-T-P-C	3-0-0-3
Course Type	PE

COURSE OBJECTIVES

- To infer the basics of internet of things and its architecture.
- To understand the architecture and networking in internet of things.
- To study the various IoT communication protocols.
- To gain insights about the fog and cloud computing in IoT framework.
- To study the real-life applications of IoT.

COURSE CONTENT

Unit-I: IoT INTRODUCTION AND FUNDAMENTALS
Basics of networking (types of networks, layered models, addressing, TCP/IP transport layer). Introduction to the architecture of wireless sensor networks, Machine-to-Machine (M2M) communication and cyber physical systems. Introduction to IoT and its comparison with M2M, WSN and CPS, IoT networking components. Addressing strategies in IoT.

Unit-II: IoT ARCHITECTURE AND NETWORKING
Introduction to IoT Sensors and their characteristics, Sensing types and their considerations. Introduction to IoT Actuators, Their types and characteristics, IoT processing topologies, their types and its importance, Data formatting, Processing topologies, IoT device design and selection considerations, Processing offloading. IoT connectivity technologies.

Unit-III: IoT COMMUNICATION TECHNOLOGIES
Introduction to cloud computing, Virtualization. Cloud Models, SLA in cloud computing, Cloud implementation, Sensor – cloud, Introduction to fog computing and its architecture, Fog computing in IoT, Application of fog computing in IoT.

Unit-IV: CLOUD AND FOG COMPUTING IN IoT
Review of RNNs, Advanced RNN architectures, Attention Models, Image Captioning, Visual QA, Visual Dialog, Spatial Transformers, Transformer Networks.

Unit-V: IoT APPLICATIONS AND DATA ANALYTICS
IoT applications in agriculture, Vehicular networks and healthcare, IoT analytics. Uses of machine learning in IoT, Advantages and challenges of ML in IoT, ML algorithms for IoT applications, Performance metrices for evaluating ML algorithms.

COURSE OUTCOMES

CO	Description	Level
CO1	Associate and classify the architecture of various communication systems.	K2
CO2	Elaborate the IoT infrastructure and data processing methodologies.	K2
CO3	Interpret the various networking protocols used in IoT.	K2
CO4	Acquire the concepts of fog and cloud computing in IoT.	K3
CO5	Illustrate the various real-life applications of IoT.	K3

TEXT BOOKS

1. Misra, Sudip, Mukherjee A., and Roy Arijit, "Introduction to IoT', Cambridge University Press, 2021.

2. Serpanos, Dimitrios, and Marilyn Wolf, "Internet-of-things (IoT) systems: architectures, algorithms, methodologies", Springer, 2017. *Deep Learning*, MIT Press.

REFERENCE BOOKS

1. Xiao, Perry, "Designing Embedded Systems and the Internet of Things (IoT) with the ARM mbed". John Wiley and Sons, 2018.

2. Hersent, Olivier, Boswarthick D., and Elloumi O., "The internet of things: Key applications and protocols" John Wiley and Sons, 2011.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PSO1	PSO2	PSO3
CO1	2	3	3	3	3	3
CO2	2	3	2	-	2	2
CO3	2	2	3	3	2	2
CO4	2	3	3	-	2	2
CO5	2	3	3	-	2	2
Score	10	14	14	6	11	11
COM	2	3	3	3	2	2

Course Code	MYPE501
Course Title	Information Theory and Coding
L-T-P-C	3-0-0-3
Course Type	PE

COURSE OBJECTIVES

- To study the fundamentals and basics of information theory.
- To demonstrate a familiarity with different coding methods.
- To apply information coding in real world applications.
- To understand the fundamentals of compression algorithms.
- To use algebraic techniques to construct efficient codes.

COURSE CONTENT

Unit-I Introduction
Memoryless Information Sources, Measure of Discrete Information, Information Entropy for a DMS (Shannon Entropy), Source Redundancy and Efficiency, Entropy of an Extended Discrete Memoryless Source, Moments and Moment Rate, Information Rate, Decision Rate, Discrete Transmission Channels.

Unit-II Discrete Transmission Channels
Probabilities and Entropies in Discrete Channels, Mutual Information and Transinformation, Relationships between Entropies, Channel Capacity Using the Noise Matrix, Shannon Capacity, Memory Sources, Finite and Homogeneous Markov Chains, Entropy of m-th Order Markov Source, Applications.

Unit-III Source Coding
What Is Coding and Why Is It Necessary?, Aim of Source Coding, Information Representation Codes, Short History, Numeral Systems, Binary Codes Used in Data Transmission, Storage or Computing, Pulse Code Modulation (PCM), Genetic Code, Coding Efficiency: Compression Ratio.

Unit-IV Lossless Compression Algorithms
Shannon-Fano Binary Coding, Huffman Algorithms, Run Length Coding (RLC), Comma Coding, Dictionary Techniques, Lempel-Ziv Type Algorithms, Arithmetic Coding, Lossy Compression in Differential Coding, Differential Pulse Code Modulation, Delta Modulation (DM), Delta-Sigma Modulation, Comparison and Applications of Digital Modulations.

Unit-V Channel Coding
Shannon Second Theorem (Noisy Channels Coding Theorem), Error Control Strategies, Classification of Error Control Codes, Representation of Binary Code Sequences, Parameters of

Detecting and Error Correcting Codes, Maximum Likelihood Decoding (MLD), Linear Block Codes, Cyclic Codes.

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Understand and analyze the fundamental concepts of coding.	K3
CO2	Illustrate the applications of discrete transmission channels.	K3
CO3	Explain the different source coding methods.	K4
CO4	Analyze the performance lossless compression algorithms.	K3
CO5	Demonstrate a familiarity with channel coding methods.	K3

TEXT BOOKS

1. Monica Borda, “Fundamentals in Information Theory and Coding”, Springer Science & Business Media, 2011.
2. Raymond W. Yeung, “Information Theory and Network Coding”, Springer Science & Business Media, 2008.
3. Yury Polyanskiy and Yihong Wu, “Information Theory: From Coding to Learning”, Cambridge University Press, 2025.

REFERENCE BOOKS

1. J. S. Chitode, “Information Theory and Coding”, Technical Publications, 2021.
2. Predrag Ivaniš and Dušan Drajić, “Information Theory and Coding—Solved Problems”, Springer International Publishing, 2017.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PSO1	PSO2	PSO3
CO1	2	2	-	3	-	3
CO2	-	-	-	3	-	3
CO3	2	2	-	-	-	-
CO4	2	2	-	-	-	-
CO5	2	2	-	-	-	3
Score	8	8	-	6	-	9
COM	2	2	-	3	-	3

Course Code	MYPE502
Course Title	Network and Database Security
L-T-P-C	3-0-0-3
Course Type	PE

COURSE OBJECTIVES

- To understand the fundamental concepts of network security, identify common attacks, and analyze security vulnerabilities in the TCP/IP protocol suite.
- To explore authentication requirements, cryptographic functions, and protocols used for secure communication and authentication in networks.
- To examine various types of malware, intrusion detection techniques, and security challenges specific to wireless networks and their defenses.
- To learn about database security challenges and implement methods such as access control, cryptographic protections, and SQL security features to safeguard data.
- To analyze database security through integrity enforcement, vulnerability assessment techniques, and methods to detect and prevent SQL injection and other attacks.

COURSE CONTENT

Unit-I Fundamentals and Threats
Overview of Network Security, Security services, attacks, Security Issues in TCP/IP suite-Sniffing, spoofing, buffer overflow, ARP poisoning, ICMP Exploits, IP address spoofing, IP fragment attack, routing exploits, UDP exploits, TCP exploits.

Unit-II Authentication Mechanisms
Authentication requirements, Authentication functions - Message Authentication Codes - Hash Functions - Security of Hash Functions and MACs - MD5 message Digest algorithm - Secure Hash Algorithm - RIPEMD - HMAC Digital Signatures, Authentication protocols-Kerberos, X.509, IP Security-AH and ESP, SSL/TLS, SSH, Web Security-HTTPS, DNS Security, Electronic Mail Security (PGP, S/MIME).

Unit-III Malware, Defense Mechanisms, and Wireless Network Security
Intruders, Viruses, Worms, Trojan horses, Distributed Denial-Of-Service (DDoS), Firewalls, IDS, Honey nets, Honey pots, introduction to wireless network security, Risks and Threats of Wireless networks, Wireless LAN Security (WEP, WPA).

Unit-IV Database Security Methods and Techniques
Database security issues, Role of databases in information systems, Access control management, Cryptographic protection, SQL security features, Statistical databases, Access control to tables/attributes/records, Triggers, Views, Data masking, Cryptographic techniques, Escaping

queries, Change Tracking, Data integrity and backups.

Unit-V Database Security Analysis
Security features using SQL, Domain/attribute/table-level integrity, Referential integrity, Monitoring tools, overview of analysis methods, SQL injection, Security scanners, Building custom tools for analysis.

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Identify and analyze common network security threats and vulnerabilities in the TCP/IP protocol suite.	K4
CO2	Explain and apply authentication functions, cryptographic hash algorithms, and secure protocols to ensure data integrity and confidentiality.	K3
CO3	Describe various types of malware, intrusion detection systems, and wireless network security threats, and evaluate appropriate defense mechanisms.	K2
CO4	Implement database security methods including access control, cryptographic protection, and SQL security features to safeguard sensitive data.	K3
CO5	Analyze database security vulnerabilities using security tools and techniques, and design solutions to prevent attacks such as SQL injection.	K4

TEXT BOOKS

1. W. Stallings, “Cryptography and Network Security: Principles and Practice”, 8th ed., Prentice Hall, 2023.
2. Aaron E. Earle, “Wireless Security Handbook”, Auerbach publications, Taylor & Francis Group, 2022.
3. A. Basta and M. Zgola, “Database Security”, 3rd Edition, Cengage Learning, 2011.
4. Ron Ben Natan, “Implementing Database Security and Auditing”, Digital Press, 2005.
5. Bhavani Thuraisingham, “Database and Applications Security”, Auerbach Publications, 2005.
6. Rose Anderson, “Security Engineering: A Guide to Building Dependable Distributed Systems”, 3rd Edition, John Wiley & Sons, 2020.

REFERENCE BOOKS

1. Yang Xiao and Yi Pan, “Security in Distributed and Networking Systems”, World Scientific, 2007.
2. Michael Gertz and Sushil Jajodia, “Handbook of Database Security: Applications and Trends”, Springer, 2008.
3. Silvana Castano, Mariagrazia Fugini, Giancarlo Martella, and Pierangela Samarati, “Database Security”, ACM Press, 1995.
4. Alfred Basta and Melissa Zgola, “Database Security”, Cengage Learning, 2011.

CO to PO/PSO Mapping

	PO1	**PO2**	**PO3**	**PSO1**	**PSO2**	**PSO3**
CO1	2	1	2	3	2	2
CO2	3	2	2	3	3	2
CO3	2	1	2	3	2	1
CO4	2	1	3	3	2	3
CO5	3	3	3	3	3	3
Score	12	8	12	15	12	11
COM	2	2	2	3	2	2

Course Code	MYPE503
Course Title	Modern Cryptography
L-T-P-C	3-0-0-3
Course Type	PE

COURSE OBJECTIVES

- To learn the design principles and security of secret-key cryptography.
- To learn the message integrity protocols and its applications.
- To understand the mathematical foundations of public-key cryptography and digital signatures.
- To apply elliptic curves for cryptographic tasks.
- To understand modern secure cryptographic applications.

COURSE CONTENT

Unit-I Secret Key Cryptography
Cryptography and Modern Cryptography, Setting of Private-Key Encryption, Principles of Modern Cryptography, Perfect Secrecy, Computational Security, Security Notions, Stream Ciphers – Pseudo-Random Generators (PRGs), Salsa and ChaCha PRGs, linear generators, cryptanalysis, Block Ciphers – DES, AES, algorithmic attacks, side-channel attacks, fault injection attacks, ideal cipher model, Chosen Plaintext Attacks (CPA) and Chosen Ciphertext Attacks (CCA).

Unit-II Message Integrity
Message Authentication Code (MAC), constructing MACs from Pseudo-Random Functions (PRFs), fully secure PRFs, converting block-wise PRF to bit-wise PRF, ANSI CBC-MAC, CMAC, PMAC, Universal Hash Functions (UHFs), MACs using UHFs, Carter-Wegman MAC, Nonce-based MACs, Merkel-Damgard paradigm, Sponge Construction, Joux's attack, HMAC, Authenticated encryption.

Unit-III Public-Key Cryptography
One-Way Trapdoor Functions (OWTFs), RSA, Diffie-Hellman (DH) Key Exchange, ElGamal Cryptosystem, Key escrow, Semantic security, Oblivious Transfer (OT) using DH, CCA security, PSA-PKCS1, Optimal Asymmetric Encryption Padding (OAEP), OAEP+, Digital Signatures – definition, secure signatures, Forking Lemma, PKCS1 signatures, signcryption, hash-based signatures, Lamport signatures, Winternitz one-time signatures, HORS.

Unit-IV Elliptic Curve Cryptography
Elliptic curves, Elliptic curve over finite fields – Montgomery and Edwards curves, Elliptic Curve Cryptography (ECC) – secp256r1, secp256k1, Curve25519, Pairing based cryptography,

signature schemes using pairings, Identity based encryption, functional encryption, multilinear maps, security of number-theoretic assumptions.

Unit-V Advanced Topics
Integer lattices, hard problems on lattices, trapdoor sampling from a lattice, lattice based signatures and encryption schemes, CRYSTALS-Kyber and Dilithium, fully homomorphic encryption, protocols for identification and login, sigma protocols, zero-knowledge proofs, authenticated key exchange, threshold cryptography, program obfuscation.

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Understand the mathematical design of secret-key cryptography and attacks	K2
CO2	Understand message integrity applications of secret-key cryptography	K2
CO3	Explain the mathematical concepts of public-key cryptography	K2
CO4	Apply the elliptic curve framework for cryptographic applications	K3
CO5	Apply modern cryptographic protocols to solve real-world problems	K3

TEXT BOOKS

1. Dan Boneh and Victor Shoup, "A Graduate Course in Applied Cryptography (v6)", available at: `https://toc.cryptobook.us/book.pdf`, 2025.
2. Jonathan Katz and Yehuda Lindell, "Introduction to Modern Cryptography", 2nd Edition, Chapman & Hall/CRC, 2012.

REFERENCE BOOKS

1. Shafi Goldwasser and Mihir Bellare, "Lecture Notes on Cryptography (2008)", available at: `https://cseweb.ucsd.edu/~mihir/papers/gb.pdf`, 2008.
2. Douglas R. Stinson and Maura B. Paterson, "Cryptography: Theory and Practice", 4th Edition, CRC Press, 2018.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PSO1	PSO2	PSO3
CO1	2	3	2	1	2	2
CO2	2	1	2	1	3	1
CO3	2	3	2	1	2	2
CO4	3	2	1	2	1	2
CO5	2	2	2	1	2	2
Score	12	11	9	6	10	9
COM	2	2	2	1	2	2

Course Code	MYPE504
Course Title	Penetration Testing and Vulnerability Assessment
L-T-P-C	3-0-0-3
Course Type	PE

COURSE OBJECTIVES

- To introduce the principles and methodology of Vulnerability Assessment and Penetration Testing (VAPT).
- To familiarize students with penetration testing phases, frameworks, and essential tools.
- To provide hands-on exposure to exploitation tools including Metasploit, and techniques for Linux and Windows exploits.
- To develop skills to identify and analyze web application security vulnerabilities and perform malware analysis.
- To enable students to conduct, document, and report a systematic security assessment ethically and professionally.

COURSE CONTENT

Unit-I Fundamentals of Vulnerability Assessment and Penetration Testing
Introduction to cybersecurity concepts; threats, vulnerabilities, and risk; phases of VAPT: planning, discovery, exploitation, reporting; legal and ethical aspects of penetration testing; standards and frameworks (e.g., OWASP, PTES, NIST).

Unit-II Penetration Testing Methodology and Tools
Reconnaissance: passive and active information gathering; scanning and enumeration; vulnerability scanners (Nessus, OpenVAS, Nikto); scripting and automation basics; reporting and documentation tools.

Unit-III Exploitation Frameworks and Operating System Exploits
Introduction to Metasploit Framework; exploitation concepts: payloads, encoders, listeners; exploiting Linux systems: privilege escalation, common misconfigurations; exploiting Windows systems: SMB exploits, token impersonation; post-exploitation strategies.

Unit-IV Web Application Security and Malware Analysis
Web application architecture and attack surface; OWASP Top 10 vulnerabilities; hands-on web exploitation with Burp Suite; introduction to malware: static and dynamic analysis; secure coding practices.

Unit-V Conducting and Reporting Security Assessments
Building a pentest engagement plan; risk scoring and vulnerability prioritization; report writing

and communication; security recommendations and remediation; ethical considerations and professional standards.

COURSE OUTCOMES

CO	Description	Level
CO1	Explain the principles, phases, and methodologies of Vulnerability Assessment and Penetration Testing within an ethical and legal framework	K2
CO2	Demonstrate practical skills in using penetration testing tools for reconnaissance, scanning, enumeration, and vulnerability identification	K3
CO3	Apply exploitation techniques with Metasploit and other frameworks to identify and exploit vulnerabilities in Linux and Windows operating systems	K3
CO4	Analyze and assess web application security vulnerabilities and perform basic malware analysis techniques	K4
CO5	Plan, conduct, document, and report a comprehensive security assessment in a systematic and professional manner	K5

TEXT BOOKS

1. Georgia Weidman, "Penetration Testing: A Hands-On Introduction to Hacking", No Starch Press, 2014.
2. Dafydd Stuttard and Marcus Pinto, "The Web Application Hacker's Handbook", Wiley, 2011.
3. Chris McNab, "Network Security Assessment", 3rd Edition, O'Reilly Media, 2017.

REFERENCE BOOKS

1. David Kennedy, Jim O'Gorman, Devon Kearns, and Mati Aharoni, "Metasploit: The Penetration Tester's Guide", No Starch Press, 2011.
2. Peter Kim, "The Hacker Playbook 3: Practical Guide to Penetration Testing", Secure Planet, 2018.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PSO1	PSO2	PSO3
CO1	2	1	2	3	2	2
CO2	3	-	3	3	2	3
CO3	3	-	3	3	2	3
CO4	2	1	3	3	3	3
CO5	3	3	3	3	3	3
Score	13	5	14	15	12	14
COM	3	2	3	3	2	3

Course Code	MYPE505
Course Title	Next-Generation Computer Networks
L-T-P-C	3-0-0-3
Course Type	PE

COURSE OBJECTIVES

- To apply the principles of networking and wireless evolution.
- To analyze communication models, routing, and security in MANETs.
- To evaluate WSN routing/data aggregation/security and trends.
- To design and simulate SDN with secure 5G/6G integration.
- To explain quantum networking and simulate secure quantum communication.

COURSE CONTENT

Unit-I Basics of Networking and Wireless Evolution
OSI/TCP-IP models, switching, protocols, addressing, routing, simulation tools. Network security basics.

Unit-II MANETs and Security Challenges
MANET types and architecture, MAC/routing protocols (AODV, DSR), security threats (wormhole, blackhole), simulations.

Unit-III WSNs, Security, and Trends
WSN architecture, MAC (S-MAC, B-MAC), routing (LEACH, PEGASIS), data aggregation, energy saving, recent trends.

Unit-IV SDN, Security, and 5G/6G Integration
SDN vs traditional networks, controllers, secure SDN, Mininet/POX tools, SDN in 5G/6G, AI-based SDN trends.

Unit-V Quantum Networks and Secure Communication
Quantum basics (Qubits, entanglement), QKD, routing, protocols, simulators (Qiskit, NetSquid), post-quantum security.

COURSE OUTCOMES

CO	Description	Level
CO1	Describe and apply core networking, protocols, and wireless evolution.	K3
CO2	Analyze routing/mobility/security in MANETs.	K4
CO3	Evaluate routing, energy, and security in WSNs.	K5
CO4	Design secure SDN and integrate with next-gen networks.	K6
CO5	Explain quantum networks and simulate secure quantum communication.	K2

TEXT BOOKS

1. Andrew S. Tanenbaum and David J. Wetherall, "Computer Networks", 5th Edition, Pearson, 2013.
2. Behrouz A. Forouzan, "Data Communications and Networking", 5th Edition, McGraw-Hill, 2013.
3. C. Siva Ram Murthy and B. S. Manoj, "Ad Hoc Wireless Networks", Pearson, 2004.
4. T. Kavitha and B. Kavitha, "Mobile Ad Hoc Networks", CRC Press, 2021.
5. Holger Karl and Andreas Willig, "Protocols for Wireless Sensor Networks", Wiley, 2005.
6. Thomas D. Nadeau and Ken Gray, "SDN: Software Defined Networks", O'Reilly, 2013.
7. Michael A. Nielsen and Isaac L. Chuang, "Quantum Computation and Quantum Information", Cambridge University Press, 2010.

REFERENCE BOOKS

1. William Stallings, "Data and Computer Communications", 10th Edition, Pearson, 2013.
2. Anand Nayyar and Er. Richa Singh, "Wireless Sensor Networks: A Review", Springer, 2020.
3. Rodney Van Meter, "Quantum Networking", Wiley-IEEE Press, 2014.
4. IBM Quantum, "Qiskit Textbook", 2021.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PSO1	PSO2	PSO3
CO1	3	3	3	3	-	-
CO2	2	3	3	3	2	-
CO3	3	2	-	-	2	-
CO4	-	2	-	-	2	-
CO5	3	2	3	3	2	-
Score	11	12	9	9	8	-
COM	3	3	3	3	2	-

Course Code	MYPE506
Course Title	Secure Computation
L-T-P-C	3-0-0-3
Course Type	PE

COURSE OBJECTIVES

- To define secure computation and its taxonomies.
- To learn secure multiparty computation protocols in honest/dishonest majority environment.
- To learn secure multiparty computation protocols in active adversary environment.
- To understand secure computation in distributed environment.
- To apply secure computation techniques to solve real-world problems.

COURSE CONTENT

Unit-I Introduction
Real-world examples, secure computation models – honest vs. dishonest majority, semi-honest vs. active adversary, static/adaptive computation, information-theoretic security, simulation-based security notions, group/ring/field overview, cryptography intro.

Unit-II Semi-honest Secure Computation
Secret Sharing, BGW construction, MPC optimizations, CDN construction, GMW construction, Yao optimizations (free XOR, point/permute), BMR construction, Oblivious Transfer, Shuffling, Sorting.

Unit-III Active Secure Computation
Verifiable secret sharing, BGW active, BH construction, hyper-invertible matrices, info checking protocol, ZK proofs, GMW compiler, Cut-and-Choose OT, Lindell-Pinkas construction.

Unit-IV Broadcast and Byzantine Agreement
Impossibility results, DS broadcast, EIG for BA, BGP for BA, multi-valued broadcast and BA.

Unit-V Advanced Topics and Applications
ABY protocols, DPF-based MPC, ORAM, privacy-preserving biometrics/genomics, secure cloud computing, private recommendation, private statistics.

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Understand secure multiparty computation and its taxonomy.	K2
CO2	Apply secure computing techniques in semi-honest setting.	K3
CO3	Apply secure computing techniques in active adversary setting.	K3
CO4	Understand the concepts of secure computation in distributed environment.	K2
CO5	Build cryptographically secure systems using secure computation.	K6

TEXT BOOKS

1. David Evans, Vladimir Kolesnikov, and Mike Rosulek, "A Pragmatic Introduction to Secure Multi-Party Computation", Now Publishers, 2018.
2. Carmit Hazay and Yehuda Lindell, "Efficient Two-Party Protocols – Techniques and Constructions", Springer, 2010.

REFERENCE BOOKS

1. Ronald Cramer, Ivan Damgård, and Jesper Buus Nielsen, "Secure Multiparty Computation and Secret Sharing", Cambridge University Press, 2015.
2. Dan Boneh and Victor Shoup, "A Graduate Course in Applied Cryptography", available at: `https://toc.cryptobook.us/book.pdf`, 2025.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PSO1	PSO2	PSO3
CO1	1	1	1	1	1	3
CO2	2	2	2	1	2	3
CO3	2	2	2	1	2	3
CO4	2	2	2	1	2	3
CO5	2	2	3	2	3	3
Score	9	9	10	6	10	15
COM	2	2	2	1	2	3

SEMESTER II

Course Code	MCPC504
Course Title	Computability and Complexity
L-T-P-C	3-1-0-4
Course Type	PC

COURSE OBJECTIVES

- To introduce abstract models of computation including Turing Machines and their capabilities
- To study decidability, undecidability, and computational intractability
- To explore time and space as computational resources and formalize complexity classes
- To understand completeness, reducibility, and circuit complexity hierarchies
- To learn about randomized, counting, and interactive complexity classes and major theorems

COURSE CONTENT

Unit-I: Computability Theory and Turing Machines
Review of abstract machines: DFA, PDA, 2DFA, Turing Machines, Encoding and enumeration of Turing Machines, Undecidability and Rice's theorem, Gödel's incompleteness theorem (via computability), Kolmogorov complexity, Blum's speedup theorem, Arithmetic and Analytic Hierarchy, Relativization.

Unit-II: Time Complexity and NP-Completeness
Time as a resource; Linear Speedup and Crossing Sequences, Time hierarchy theorems, Classes P, NP, EXP, Polynomial-time reductions, Cook–Levin Theorem, Polynomial Hierarchy and Ladner's Theorem, P vs NP and the relativization barrier.

Unit-III: Space Complexity and Reachability
Space as a resource; Classes L, NL, PSPACE, Savitch's Theorem, NL-Completeness, Inductive counting and Immerman–Szelepcsényi Theorem (NL = coNL), Expander graphs, SL = L, Space hierarchy and PSPACE-completeness.

Unit-IV: Circuit and Randomized Complexity
Boolean circuits, AC^0, NC, P/poly, Circuit Hierarchy Theorem, Karp–Lipton Theorem, Adleman's Theorem, Parity not in AC^0, Randomized classes: RP, BPP, ZPP, Polynomial Identity Testing, RNC algorithms, Isolation Lemma, BPP in PH.

Unit-V: Counting Complexity and Interactive Proofs
Counting class $\#P$ and $\#P$-Completeness (e.g., Permanent), Classes PP, $\mathsf{ParityP}$, Toda's Theorem ($\mathsf{PH} \subseteq \mathsf{P}\#\mathsf{P}$), Communication complexity, monotone circuit lower bounds, Introduction to Interactive Proofs: IP, $\#3\mathrm{SAT} \in \mathsf{IP}$, public vs private coins.

COURSE OUTCOMES

CO	Description	Level
CO1	Analyze various abstract computation models and their power	K4
CO2	Determine decidability and undecidability using theoretical tools	K3
CO3	Classify problems using time and space complexity classes	K4
CO4	Apply reduction techniques to prove NP-completeness and other completeness results	K3
CO5	Explain and evaluate randomized, counting, and interactive proof complexity classes	K5

TEXT BOOKS

1. Sanjeev Arora and Boaz Barak, *Computational Complexity: A Modern Approach*, Cambridge University Press, 2009.

2. Michael Sipser, *Introduction to the Theory of Computation*, Cengage Learning, 2012.

REFERENCE BOOKS

1. Dexter Kozen, *Automata and Computability*, Springer, 1997.

2. Sanjeev Arora and Boaz Barak, *Theory of Computational Complexity*, Cambridge University Press, 2009.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PSO1	PSO2	PSO3
CO1	3	2	3	3	2	2
CO2	3	2	2	2	2	2
CO3	3	3	3	3	3	2
CO4	3	2	3	3	3	3
CO5	2	3	3	2	3	3
Score	14	12	14	13	13	12
COM	3	2	3	3	3	2

Course Code	MCPC505
Course Title	Distributed Systems
L-T-P-C	3-0-2-4
Course Type	PC

COURSE OBJECTIVES

- To introduce the fundamentals of distributed systems, including architecture, models, interprocess communication, and resource sharing.
- To understand distributed object communication, operating system support for distributed processes, and foundational security mechanisms in distributed environments.
- To explore the role of the operating system in managing distributed file systems, naming services, synchronization, coordination, and peer-to-peer systems.
- To examine the principles of distributed transactions, concurrency control, atomic commit protocols, recovery mechanisms, and replication.
- To analyze and apply distributed algorithms for coordination, mutual exclusion, and leader election in both synchronous and asynchronous network models.

COURSE CONTENT

Unit-I: Basic Concepts
Definition of a distributed systems, Examples, Resource sharing and the Web, Challenges, System models, Architectural and fundamental models, Networking Interprocess communication, External data representation and marshalling, Client-server and Group communication.

Unit-II: Distributed Objects and Process
Distributed objects and remote invocation, Communication between distributed objects, Remote procedure call, Events and notifications - The operating system layer, Protection, Processes and Threads, Communication and invocation, OS Architecture. Security techniques, Cryptographic algorithms, Access control, Digital signatures, Cryptography pragmatics, Needham-Schroeder, Kerberos, Securing electronics transaction, IEEE 802.11 WiFi.

Unit-III: Operating System Issues
Distributed file systems - Name services, Domain name system, Directory and discovery services, Peer to peer systems, Napster file sharing system, Peer to peer middleware routing overlays – Clocks, Events and process states Clock Synchronization - Logical clocks Global states - Distributed debugging - Distributed mutual exclusion - Elections - Multicast communication.

Unit-IV: Distributed Transaction Processing
Transactions - Nested transactions - Locks - Optimistic concurrency control - Timestamp ordering - Flat and nested distributed transactions - Atomic commit protocols - Concurrency control

in distributed transactions - Distributed deadlocks - Transaction recovery - Overview of replication, Distributed shared memory and Web services.

Unit-V: Distributed Algorithms
Synchronous network model - Algorithms: leader election, maximal independent set - Asynchronous system model: I/O automata, operations on automata, fairness - Asynchronous shared memory model - Mutual exclusion: model, the problem, stronger conditions, lockout-free mutual exclusion algorithms, lower bound on the number of registers - Asynchronous network model - Asynchronous network algorithms: leader election in a ring and an arbitrary network.

LAB CONTENT

The experiments may be performed as per the course contents mentioned above.

COURSE OUTCOMES

CO	Description	Level
CO1	Explain the fundamental concepts, architectural models, and communication mechanisms in distributed systems.	K2
CO2	Describe distributed object communication, operating system roles, and implement basic security techniques for distributed environments.	K3
CO3	Analyze the challenges of distributed file systems, naming services, synchronization, and peer-to-peer systems, and explain their OS-level management.	K4
CO4	Apply concepts of distributed transactions, concurrency control, and recovery techniques to ensure consistency and fault tolerance.	K3
CO5	Design and evaluate distributed algorithms for mutual exclusion, leader election, and coordination in various network models.	K5

TEXT BOOKS

1. George Coulouris, Jean Dollimore, and Tim Kindberg, *Distributed Systems Concepts and Design*, 5th ed., Pearson Education, 2017.

2. Andrew S. Tanenbaum, Maartenvan Steen, *Distributed Systems Principles and Paradigms*, 2nd ed., Pearson Education, 2016.

REFERENCE BOOKS

1. Ranjit Aneesh, *Building Distributed Systems: Designing scalable architectures with microservices, event-driven patterns, and robust deployment strategies*, 1st ed., BPB Publi-

cations, 2025.

2. Nancy A. Lynch, *Distributed Algorithms*, Hardcourt Asia Pvt. Ltd., Morgan Kaufmann, 2000.

CO to PO/PSO Mapping

	PO1	**PO2**	**PO3**	**PSO1**	**PSO2**	**PSO3**
CO1	2	2	2	2	2	2
CO2	2	1	1	2	1	2
CO3	2	2	3	2	2	1
CO4	2	2	3	3	3	2
CO5	2	2	2	3	3	3
Score	10	9	11	12	11	10
COM	2	2	2	2	2	2

Course Code	MDPE507
Course Title	Big Data Analytics
L-T-P-C	3-0-0-3
Course Type	PE

COURSE OBJECTIVES

- To familiarize students with big data analysis as a tool for analysing large complex dataset.
- To learn to use various techniques for mining data stream.
- To understand the applications using Map Reduce Concepts.
- To provide hands on Hodoop Eco-system.
- To introduce programming tools PIG & HIVE in Hadoop echo system.

COURSE CONTENT

Unit-I: NoSQL Database
NoSQL Database: NoSQL Databases - Schema less Models, Increasing Flexibility for Data Manipulation-Key Value Stores, Document Stores, Tabular Stores, Object Data Stores - Graph Databases, Big data for twitter, Big data for E-Commerce blogs.

Unit-II: Data Visualization
Basics of data visualization, Principles used for visualization, Lie factor, Category based visualization, Visualization tools and techniques, Outlier detection using visualization, Visualization based data analysis techniques.

Unit-III: Big Data
Best Practices for Big data Analytics - Big data characteristics - Big Data Use Cases, Characteristics of Big Data Applications, Big Data Modelling, HDFS performance and tuning, Map reduce algorithm.

Unit-IV: Hadoop and Hive
Hadoop Eco system Pig : Introduction to PIG, Execution Modes of Pig, Grunt, Pig Latin, User Defined Functions, Data Processing operators. Hive : Hive Shell, Hive Services, HiveQL, Tables, Querying Data and User Defined Functions. Hbase : HBasics, Concepts, Clients, Example, Spark.

Unit-V: Mining Data Streams
Introduction to Streams Concepts, Stream Data Model and Architecture - Sampling Data in a Stream, Filtering Streams, Counting Distinct Elements in a Stream –Real time Analytics Platform (RTAP) applications, Case Studies, Real Time Sentiment Analysis- Stock Market Predictions.

COURSE OUTCOMES

CO	Description	Level
CO1	Process data in Big Data platforms.	K1
CO2	Explore the big data analytics techniques for business applications.	K2
CO3	Analyse Map Reduce technologies in big data analytics.	K3
CO4	Develop Big Data solutions using Hadoop Eco System.	K4
CO5	Design efficient algorithms for stream data mining on big data platform.	K5

TEXT BOOKS

1. David Loshin, ”Big Data Analytics: From Strategic Planning to Enterprise Integration with Tools, Techniques, NoSQL, and Graph”, 2013.

2. Bart Baesens, ”Analytics in a Big Data World: The Essential Guide to Data Science and its Applications”, Wiley Publishers, 2015.

REFERENCE BOOKS

1. Kim H. Pries and Robert Dunnigan, ”Big Data Analytics: A Practical Guide for Managers ”CRC Press, 2015.

2. EMC Education Services, ”Data Science and Big Data Analytics: Discovering, Analyzing, Visualizing and Presenting Data”, Wiley publishers, 2015.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PSO1	PSO2	PSO3
CO1	3	1	2	3	2	2
CO2	3	2	2	3	3	2
CO3	3	2	3	3	3	3
CO4	3	2	3	3	2	3
CO5	3	3	3	3	3	3
Score	15	10	13	15	13	13
COM	3	2	3	3	3	3

Course Code	MDPE508
Course Title	Computer Vision Algorithms
L-T-P-C	3-0-0-3
Course Type	PE

COURSE OBJECTIVES

- To learn the fundamental concepts in computer vision.
- To learn the basics of image feature detection and segmentation.
- To review and use deep learning models for computer vision.
- To make use of deep learning models for video analysis.
- To learn generative models and their application in computer vision tasks.

COURSE CONTENT

Unit-I: Introduction to Computer Vision
A brief history of Computer Vision, Image Formation, Geometric Primitives and Transformations, Image Processing: Point Operators, Linear Filtering, Correlation, Convolution, Fourier Transforms, Pyramid and Wavelets, Geometric Transformations.

Unit-II: Feature Detection and Segmentation in Images
Point, Edge, and Line Detection, Feature Extraction: SIFT, SURF, LBP, and Texture analysis, Visual Matching: Bag-of-words, and VLAD, Segmentation: Active contours, Split and merge, Mean shift and mode finding, Normalized cuts, Graph cuts and energy based methods, Optical Flow for Videos.

Unit-III: CNNs in Computer Vision
Review of Deep Learning: FFNN and Backpropagation, Introduction to CNNs, Evolution of CNN Architectures, Visualization of Kernels: CAM, Grad-CAM, CNNs for Recognition and Verification: Siamese Networks, Triplet Loss, Contrastive Loss, Ranking Loss, CNNs for Detection: R-CNN, Fast R-CNN, Faster R-CNN, YOLO, SSD, RetinaNet, CNNs for Segmentation: FCN, SegNet, U-Net, Mask RCNN.

Unit-IV: RNNs and Attention Models
Review of RNNs, Advanced RNN architectures, Attention Models, Image Captioning, Visual QA, Visual Dialog, Spatial Transformers, Transformer Networks.

Unit-V: Deep Generative Models and Applications in Computer Vision
GANs, VAEs, Pixel-RNNs; Applications: Image Editing, Inpainting, Super-resolution, 3D Object Generation, Security; Variants: Cycle GANs, Progressive GANs, Stack GANs, Pix2Pix; Recent Trends: Zero-shot, One-shot, Few-shot Learning.

COURSE OUTCOMES

CO	Description	Level
CO1	Explain the fundamentals of computer vision.	K2
CO2	Apply various feature detection techniques on images.	K3
CO3	Examine the outcomes of various CNN architectures.	K4
CO4	Make use of Attention in RNN and CNN models.	K3
CO5	Develop advanced CV algorithms using Generative Networks.	K5

TEXT BOOKS

1. Szeliski, R. (2010). *Computer Vision: Algorithms and Applications*, Springer.
2. Goodfellow, I., Bengio, Y., Courville, A. (2016). *Deep Learning*, MIT Press.

REFERENCE BOOKS

1. Prince, S. J. D. (2012). *Computer Vision: Models, Learning, and Inference*, Cambridge University Press.
2. Nielsen, M. (2016). *Neural Networks and Deep Learning*, Springer.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PSO1	PSO2	PSO3
CO1	3	2	1	2	2	1
CO2	2	3	2	2	2	2
CO3	2	3	3	3	3	2
CO4	2	3	3	3	3	3
CO5	2	2	3	3	3	3
Score	11	13	12	13	13	11
COM	2	3	2	3	3	2

Course Code	MDPE509
Course Title	Natural Language Processing
L-T-P-C	3-0-0-3
Course Type	PE

COURSE OBJECTIVES

- To learn core NLP concepts, preprocessing, and traditional statistical models for text analysis.
- To understand word embedding and sequence models for semantic text processing.
- To learn transformer architectures and fine-tuning pre-trained models for advanced NLP tasks.
- To apply NLP to real-world tasks like translation, question answering, and summarization.
- To explore cutting-edge NLP trends, model deployment, and research methodologies for practical applications.

COURSE CONTENT

Unit-I NLP Fundamentals
Introduction to NLP: applications, challenges, pipeline, text preprocessing: tokenization, stemming, lemmatization, stop-word removal, linguistic concepts: POS tagging, NER, dependency parsing, traditional models: bag-of-words, TF-IDF, naive Bayes, tools: NLTK, spaCy, scikit-learn, evaluation metrics: precision, recall, F1, BLEU, ROUGE.

Unit-II Word Embeddings & Sequence Models
Word embeddings: Word2Vec, GloVe, fastText, contextual vs. non-contextual, sequence models: RNNs, LSTMs, GRUs, bidirectional RNNs, applications: sentiment analysis, language modeling, text classification, frameworks: TensorFlow, PyTorch basics, model training.

Unit-III Transformers & Pre-trained Models
Attention mechanisms: self-attention, multi-head attention, scaled dot-product, transformer architecture: encoder-decoder, positional encodings, pre-trained models: BERT, GPT, RoBERTa, T5, fine-tuning techniques, tools: Hugging Face transformers, datasets, pipelines.

Unit-IV Advanced NLP Applications
Machine translation: Seq2Seq, transformer-based translation, question answering: extractive, generative, RAG, text summarization: extractive, abstractive, BART, PEGASUS, dialogue systems: chatbots, intent recognition, dialogue management, ethics: bias, fairness, societal impact.

Unit-V Research & Deployment
Emerging trends: large language models, few-shot/zero-shot learning, model compression, pruning, distillation, deployment: APIs with FastAPI/Flask, cloud platforms like AWS/GCP, interpretability: SHAP, LIME, attention visualization, research methodologies: analyzing research papers, case studies.

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Preprocess text and apply traditional NLP models	K2
CO2	Implement embeddings and sequence models	K3
CO3	Fine-tune transformers for advanced NLP tasks	K4
CO4	Develop systems for translation, QA, and summarization	K4
CO5	Deploy models and analyze NLP research trends	K4

TEXT BOOKS

1. Jurafsky, Dan and Martin, James, "Speech and Language Processing", 3rd Edition, Pearson, 2024.
2. Tunstall, Lewis, et al., "Natural Language Processing with Transformers", O'Reilly Media, 2022.
3. Ganegedara, Thushan, "Natural Language Processing with TensorFlow", 2nd Edition, Packt Publishing, 2022.

REFERENCE BOOKS

1. Allen, James, "Natural Language Understanding", 2nd Edition, Benjamin Cummings, 2002.
2. Charniack, Eugene, "Statistical Language Learning", MIT Press, 1996.
3. Vajjala, Sowmya, et al., "Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems", O'Reilly Media, 2020.
4. Manning, Christopher and Heinrich Schutze, "Foundations of Statistical Natural Language Processing", MIT Press, 1999.
5. Raaijmakers, Stephan, "Deep Learning for Natural Language Processing", Manning Publications, 2019.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PSO1	PSO2	PSO3
CO1	2	1	2	3	1	2
CO2	2	2	3	3	2	2
CO3	3	2	3	3	3	3
CO4	3	3	3	3	2	3
CO5	3	3	3	3	3	3
Score	13	11	14	15	11	13
COM	3	2	3	3	2	3

Course Code	MDPE510
Course Title	Advanced Deep Networks
L-T-P-C	3-0-0-3
Course Type	PE

COURSE OBJECTIVES

- To provide a deep understanding of advanced neural architectures and their mathematical foundation.
- To explore modern developments in deep neural networks including CNNs, RNNs, Transformers, and GNNs.
- To introduce optimization strategies, regularization, and tuning methods used in training deep networks.
- To apply neural networks in edge-AI, real-time systems, and embedded hardware platforms.
- To expose students to research trends in neural architecture search, explainable AI, and tinyML.

COURSE CONTENT

Unit-I Fundamentals & Architectures
Backpropagation, activation functions, loss functions, residual networks (ResNet), DenseNets, capsule networks, SqueezeNet, MobileNet; initialization and optimization – weight initialization, batch norm, gradient clipping.

Unit-II Recurrent & Attention-based Networks
Recurrent neural networks, LSTM, GRU, bidirectional RNNs; attention mechanisms, self-attention, multi-head attention, sequence-to-sequence models; transformers architecture, applications in NLP and vision.

Unit-III Graph Neural Networks & Autoencoders
Autoencoders, denoising, sparse, variational autoencoders; graph neural networks, message passing, GCN, GAT, contrastive learning in neural architectures.

Unit-IV Embedded & Efficient Neural Networks
Model compression, quantization, pruning, knowledge distillation; tinyML and edge AI deploying networks on microcontrollers; ONNX, TensorFlow Lite, Edge TPU, and embedded AI SDKs.

Unit-V Explainability, NAS & Recent Trends
Explainable AI, saliency maps, Grad-CAM, LIME, SHAP; neural architecture search, NASNet,

EfficientNet, reinforcement learning-based NAS; recent trends large language models (LLMs), multimodal networks, diffusion models overview.

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Apply advanced deep learning architectures for real-world problems	K3
CO2	Design and implement sequence and attention-based models for structured data	K3
CO3	Develop interpretable and explainable models using modern XAI tools	K4
CO4	Optimize and deploy neural models on edge and embedded devices	K4
CO5	Critically analyze and contribute to current research trends in neural network design	K5

TEXT BOOKS

1. Ian Goodfellow, Yoshua Bengio, and Aaron Courville, "Deep Learning", MIT Press, 2016.
2. Charu C. Aggarwal, "Neural Networks and Deep Learning: A Textbook", Springer, 2018.

REFERENCE BOOKS

1. Sebastian Raschka, Yuxi Liu, "Machine Learning with PyTorch and Scikit-Learn", Packt, 2022.
2. TinyML Foundation, Pete Warden et al., "TinyML: Machine Learning with TensorFlow Lite on Arduino and Ultra-Low Power Microcontrollers", O'Reilly, 2019.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PSO1	PSO2	PSO3
CO1	3	1	2	3	2	2
CO2	3	2	2	3	3	2
CO3	3	2	3	3	3	3
CO4	3	2	3	3	2	3
CO5	3	3	3	3	3	3
Score	15	10	13	15	13	13
COM	3	2	3	3	3	3

Course Code	MDPE511
Course Title	Applied Accelerated Artificial Intelligence
L-T-P-C	3-0-0-3
Course Type	PE

COURSE OBJECTIVES

- To study the fundamentals of hardware and software components behind modern AI infrastructure.
- To enable students to understand the automation tools infrastructure to deploy AI based services on a cluster.
- To enable students to utilize various frameworks used for accelerating deep learning and apply optimization techniques for efficient model training.
- To study the fundamentals of distributed artificial intelligence computing and its deployment using various neural network inference.
- To explore various distributed frameworks to execute tools and methodologies for scalable data analytics and explore real-time practical applications.

COURSE CONTENT

Unit-I AI Infrastructure: Hardware, Software, and Orchestration
Introduction to AI system hardware: CPU, RAM, GPU, interconnects, storage, network controller; introduction to AI accelerators GPUs; introduction to system software operating system, virtualization, cloud. Introduction to containers and IDE using Jupyter; scheduling and resource management; introduction to schedulers and orchestration tools.

Unit-II DeepOps: Scalable AI Deployment
DeepOps: deep-dive into Kubernetes with deployment of various AI-based services; design principles for building high performance compute clusters for AI; implementation details for building high performance compute clusters for AI.

Unit-III Deep Learning Frameworks and Optimization
Frameworks for accelerated deep learning workloads - PyTorch, accelerated PyTorch; frameworks for accelerated deep learning workloads - TensorFlow, accelerated TensorFlow; optimizing deep learning training: automated mixed precision; optimizing deep learning training: transfer learning.

Unit-IV Distributed AI and Accelerated Deployment
Fundamentals of distributed AI computing: multi-GPU and multi-node implementation (MPI, NCCL, RDMA); distributed AI computing: Horovod; challenges with distributed deep learning training convergence; fundamentals of accelerating deployment; accelerating neural network

inference in PyTorch and TensorFlow; accelerated data analytics; accelerated machine learning.

Unit-V Scalable Data Analytics and Applied AI
Scale out with DASK; web visualizations to GPU accelerated crossfiltering; accelerated ETL pipeline with Spark; applied AI: smart city (intelligent video analytics); applied AI: healthcare (federated learning, AI assisted annotation).

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Identify and describe the core hardware and software elements (e.g., CPU, GPU, OS, virtualization, containers) and their roles in setting up an AI development environment	K3
CO2	Deploy and manage AI-based services effectively using Kubernetes within the context of high-performance compute clusters	K2
CO3	Implement and optimize deep learning models using PyTorch and TensorFlow, incorporating techniques like mixed precision and transfer learning	K4
CO4	Explain and apply methods for distributed AI training across multiple GPUs/nodes and accelerate neural network inference for deployment	K2
CO5	Implement scalable data analytics solutions using tools like DASK and analyze various real-world applications of AI	K2

TEXT BOOKS

1. Charles Petzold, “Code: The Hidden Language of Computer Hardware and Software”, 2nd Edition, Microsoft Press, 2022.
2. Jon Krohn, Grant Beyleveld, Aglaé Bassens, “Deep Learning Illustrated”, Addison-Wesley, 2019.
3. Neylson Crepalde, “Big Data on Kubernetes: A Practical Guide to Building Efficient and Scalable Data Solutions”, Packt Publishing, 2024.
4. K. Bonawitz, P. Kairouz, B. McMahan, D. Ramage, “Federated Learning and Privacy: Building Privacy-Preserving Systems for Machine Learning and Data Science on Decentralized Data”, ACM New York, 2021.
5. Bill Chambers, Matei Zaharia, “Spark: The Definitive Guide”, O’Reilly Media, 2018.

REFERENCE BOOKS

1. Ashok Srirama, Sukirti Gupta, “Kubernetes for Generative AI Solutions”, 1st Edition, Packt Publishing, 2025.

2. Ian Pointer, “Programming PyTorch for Deep Learning: Creating and Deploying Deep Learning Applications”, O’Reilly, 2019.

3. Aurélien Géron, “Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems”, 3rd Edition, Shroff / O’Reilly, 2022.

CO to PO/PSO Mapping

	PO1	**PO2**	**PO3**	**PSO1**	**PSO2**	**PSO3**
CO1	3	1	3	3	1	2
CO2	3	1	3	3	1	3
CO3	3	1	3	3	2	3
CO4	3	1	3	3	2	3
CO5	3	1	3	3	2	3
Score	15	5	15	15	8	14
COM	3	1	3	3	2	3

Course Code	MDPE512
Course Title	Large Language Models
L-T-P-C	3-0-0-3
Course Type	PE

COURSE OBJECTIVES

- To introduce foundational concepts behind modern large-scale language models.
- To explore the architectures, training methods, and optimization of Transformer-based LLMs.
- To examine pre-training and fine-tuning paradigms in context of real-world tasks.
- To explore ethical considerations, biases, and safety in LLM deployment.
- To provide hands-on experience with building, fine-tuning, and deploying LLMs using open-source tools.

COURSE CONTENT

Unit-I Foundations of Language Modeling
Basics of language models, n-grams to neural LMs; word embeddings, Word2Vec, GloVe, FastText; sequence modeling limitations, motivation for attention, Transformer architecture – scaled dot-product attention, multi-head attention, positional encoding.

Unit-II Pre-training Paradigms
Self-supervised learning, MLM (BERT), CLM (GPT), T5-style pretraining; unsupervised objectives, next token prediction, masked span prediction; tokenization, byte-pair encoding (BPE), SentencePiece; datasets C4, The Pile, BooksCorpus, WebText.

Unit-III Fine-tuning and Prompt Engineering
Fine-tuning – classification, QA, summarization using BERT/GPT/LLAMA; low-rank adaptation (LoRA), PEFT, adapter tuning; prompt engineering – zero-shot, few-shot, chain-of-thought (CoT), role-based prompting; instruction tuning, RLHF (reinforcement learning with human feedback).

Unit-IV LLM Applications & Tools
Applications – chatbots, code generation, text summarization, semantic search; toolkits – HuggingFace Transformers, LangChain, OpenAI API, Sentence Transformers; deployment – quantization (INT8, GPTQ), ONNX, NVIDIA TensorRT; LLMs on edge – DistilBERT, TinyBERT, Whisper for ASR.

Unit-V Ethics, Bias, Safety, and Recent Trends
LLM biases, toxicity, hallucinations, detection and mitigation; ethical use guidelines (OpenAI,

Google, Anthropic); safety frameworks – Constitutional AI, Guardrails, Moderation APIs; recent trends – Llama 3, GPT-4o, Claude, Gemini; open-weight models – Mistral, Falcon, BLOOM, Mixtral, Orca, Qwen, Phi.

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Understand the theory and implementation of Transformer-based large language models	K2
CO2	Design and fine-tune LLMs for a variety of NLP tasks	K3
CO3	Apply prompt engineering and RAG techniques to improve LLM outputs	K4
CO4	Optimize, quantize and deploy LLMs on local or edge platforms	K3
CO5	Critically assess LLM performance in terms of safety, ethics, and bias	K4

TEXT BOOKS

1. Sebastian Ruder et al., "Transformers and Attention for NLP", Online Book/Resources, 2021.
2. HuggingFace Team, "The Transformers Book (Early Access)", 2024.

REFERENCE BOOKS

1. Andriy Burkov, "Machine Learning Engineering", True Shelf, 2020.
2. OpenAI Technical Reports – GPT-3, GPT-4, InstructGPT, GPT-4o (Whitepapers), 2024.
3. Thomas Wolf et al., "HuggingFace Course", `https://huggingface.co/learn`, 2023
4. Anthropic and Google DeepMind papers on safety, alignment, and RLHF, 2022.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PSO1	PSO2	PSO3
CO1	3	1	2	3	2	2
CO2	3	2	2	3	3	2
CO3	3	2	3	3	3	3
CO4	3	2	3	3	2	3
CO5	3	3	3	3	3	3
Score	15	10	13	15	13	13
COM	3	2	3	3	3	3

Course Code	MYPE507
Course Title	Malware Analysis and Intrusion Detection
L-T-P-C	3-0-0-3
Course Type	PE

COURSE OBJECTIVES

- To introduce the fundamentals of malware and its classification.
- To develop skills in static and dynamic malware analysis techniques.
- To understand and implement intrusion detection systems (IDS).
- To gain hands-on experience with tools like Snort, Suricata, and memory forensics frameworks.
- To analyze attack patterns and apply detection and prevention strategies.

COURSE CONTENT

Unit-I Introduction to Malware and Intrusion Detection Systems
Malware types: viruses, worms, trojans, ransomware, rootkits; malware lifecycle; IDS vs. IPS; signature-based and anomaly-based IDS; intro to Snort and Suricata.

Unit-II Static Malware Analysis
Disassembly basics, PE and ELF formats, strings, metadata, static indicators, PEiD, IDA Pro, Ghidra.

Unit-III Dynamic Malware Analysis
Sandboxing, instrumentation, behavioral monitoring, malware lab setup, Cuckoo Sandbox, Process Monitor, Wireshark.

Unit-IV Intrusion Detection Techniques and Tools
Network-based vs. host-based IDS, Snort rule writing, Suricata features, evasion techniques, performance tuning.

Unit-V Memory Forensics and Incident Response
Memory forensics intro, acquisition tools, Volatility framework, rootkit detection, IR lifecycle, documentation and reporting.

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Explain the categories, characteristics, and lifecycle of malware and intrusion detection systems.	K2
CO2	Perform static and dynamic malware analysis using industry tools.	K3
CO3	Implement, configure, and fine-tune intrusion detection systems like Snort and Suricata.	K3
CO4	Apply memory forensics techniques to detect advanced malware and rootkits.	K4
CO5	Plan and document a full incident response, including detection, analysis, and reporting.	K5

TEXT BOOKS

1. Michael Sikorski and Andrew Honig, "Practical Malware Analysis", No Starch Press, 2012.
2. Stephen Northcutt and Judy Novak, "Network Intrusion Detection", New Riders, 2002.
3. Cameron Malin, Eoghan Casey, and James M. Aquilina, "Malware Forensics", Syngress, 2008.
4. Mahadev Thukaram and Dharmendra T., "Advanced Malware Analysis and Intelligence", BPB Publications, 2025.
5. Mayank Swarnkar and Shyam Singh Rajput, "Artificial Intelligence for Intrusion Detection Systems", CRC Press / Taylor & Francis, 2025.

REFERENCE BOOKS

1. Michael Hale Ligh, Andrew Case, Jamie Levy, and AAron Walters, "The Art of Memory Forensics", Wiley, 2014.
2. Rafeeq Ur Rehman, "Intrusion Detection with Snort", Prentice Hall, 2003.
3. Dimitris Gritzalis, Kim-Kwang Raymond Choo, and Constantinos Patsakis, "Malware: Handbook of Prevention and Detection", Springer, 2024.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PSO1	PSO2	PSO3
CO1	2	1	2	3	2	2
CO2	3	-	3	3	2	3
CO3	3	-	3	3	2	3
CO4	3	-	3	3	3	3
CO5	3	3	3	3	3	3
Score	14	4	14	15	12	14
COM	3	2	3	3	2	3

Course Code	MYPE508
Course Title	Mobile Security
L-T-P-C	3-0-0-3
Course Type	PE

COURSE OBJECTIVES

- To introduce fundamental concepts and challenges of mobile application security.
- To study the security architectures of various mobile platforms like Android and iOS.
- To understand threats, vulnerabilities, and protection mechanisms in platform-specific and cross-platform apps.
- To develop practical skills in performing penetration testing on mobile applications using industry-standard tools and methodologies.
- To provide hands-on exposure to secure development practices across Android, iOS, and hybrid frameworks.

COURSE CONTENT

Unit-I Introduction to Mobile Application Security
Mobile app security basics, threats, OS security principles, sandboxing, permissions, secure coding, mobile breach case studies.

Unit-II iOS Application Security
iOS architecture, sandboxing, keychain, provisioning, TLS, iOS malware and exploits, secure app practices, static/dynamic analysis.

Unit-III Android Application Security
Rooting risks, component-based threats, permissions, HTTPS, SELinux, Safetynet, App Defense Alliance, reverse engineering, obfuscation.

Unit-IV Mobile Application Penetration Testing
OWASP mobile top vulnerabilities, testing tools (MobSF, Frida, Burp), static/dynamic analysis, API testing, traffic interception.

Unit-V Cross-Platform Application Security
Frameworks: Cordova, React Native; plugin vulnerabilities, JS injection, secure communication/storage, testing hybrid apps, secure design.

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Understand the foundational concepts of mobile application security, including mobile OS architecture, app sandboxing, permissions, and secure coding practices.	K2
CO2	Analyze the iOS security model, identify common vulnerabilities in iOS apps, and apply secure development and testing techniques.	K4
CO3	Evaluate Android application threats, perform reverse engineering, and implement advanced protection mechanisms like obfuscation and anti-debugging.	K5
CO4	Perform mobile app penetration testing using open-source tools, identify OWASP Mobile top vulnerabilities, and mitigate security flaws via analysis.	K3
CO5	Design secure cross-platform applications, detect hybrid-specific threats, and implement secure storage and communication mechanisms in hybrid frameworks.	K6

TEXT BOOKS

1. Aditya Gupta and Manish Gupta, “Mobile Application Security”, 1st Edition, McGraw Hill, 2016.
2. Jonathan Zdziarski, “Hacking and Securing iOS Applications”, 1st Edition, O’Reilly Media, 2012.
3. Narayan Ramachandran and Abhinav Yadav, “Android Security Internals”, 1st Edition, No Starch Press, 2014.
4. Dominic Chell Hunt and Jamie D. Dewey, “Mobile Application Penetration Testing”, 1st Edition, Wiley, 2015.
5. Daniel L. P. Real, “Hands-On Mobile App Testing”, 1st Edition, Addison-Wesley, 2016.

REFERENCE BOOKS

1. Steven M. Furnell, “Mobile Device Security: A Practical Guide for Managers”, Chandos Publishing, 2010.
2. B. S. Ravi, “Android Application Security Essentials”, Packt Publishing, 2014.
3. Abhijit Hoque, “Hacking Android”, Packt Publishing, 2017.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PSO1	PSO2	PSO3
CO1	3	3	2	3	2	-
CO2	2	3	3	3	2	-
CO3	3	2	3	2	3	-
CO4	2	2	2	2	2	2
CO5	3	3	3	3	3	2
Score	13	13	13	13	12	4
COM	3	3	3	3	2	2

Course Code	MYPE509
Course Title	Blockchain Technologies
L-T-P-C	3-0-0-3
Course Type	PE

COURSE OBJECTIVES

- To demonstrate a familiarity with the concepts related to blockchain technology.
- To apply the knowledge of distributed systems and cryptography to design decentralized applications.
- To learn the technologies related to Blockchain 1.0, 2.0 and cryptocurrencies.
- To understand the concepts of Blockchain 3.0 to 4.0.
- To analyze the privacy and security issues related to blockchains.

COURSE CONTENT

Unit-I Introduction
Need for distributed record keeping, modeling faults and adversaries, Byzantine Generals problem, consensus algorithms and their scalability problems, why Nakamoto came up with blockchain-based cryptocurrency, technologies borrowed in blockchain – hash pointers, consensus, byzantine fault-tolerant distributed computing, digital cash.

Unit-II Distributed Computing and Crypto Primitives
Atomic broadcast, consensus, Byzantine models of fault tolerance, consensus protocols – proof of work (PoW), scalability aspects of blockchain consensus protocols, permissioned blockchains – design goals, consensus protocols for permissioned blockchains, hash functions, puzzle-friendly hash, collision-resistant hash, digital signatures, public key crypto, verifiable random functions, zero-knowledge systems.

Unit-III Blockchain 1.0 and 2.0
Bitcoin blockchain, the challenges and solutions, proof of work, proof of stake, alternatives to Bitcoin consensus, Bitcoin scripting language and their use, Ethereum and smart contracts, the Turing completeness of smart contract languages and verification challenges, using smart contracts to enforce legal contracts, comparing Bitcoin scripting vs. Ethereum smart contracts.

Unit-IV Blockchain 3.0 to 5.0
Hyperledger Fabric, DeFi, NFT, the plug and play platform and mechanisms in permissioned blockchain, blockchain and IoT integration, AI and blockchain, automation and smart industry, decentralized web, integration with AR, VR, and metaverse, ethics, privacy, and regulation.

Unit-V Privacy, Security Issues
Pseudo-anonymity vs. anonymity, Zcash and zk-SNARKS for anonymity preservation, attacks on blockchains such as Sybil attacks, selfish mining, 51% attacks, advent of Algorand and sharding-based consensus algorithms to prevent these.

COURSE OUTCOMES

CO	Description	Level
CO1	Understand the design principles of Blockchain	K2
CO2	Apply the concepts of distributed computing and cryptography in Blockchain	K3
CO3	Explain technologies related to Blockchain 1.0, 2.0, and cryptocurrencies	K2
CO4	Understand the concepts of Blockchain 3.0 to 5.0	K2
CO5	Analyze the privacy and security issues of the Blockchains	K4

TEXT BOOKS

1. Roger Wattenhofer, "Blockchain Science: Distributed Ledger Technology", Independently Published, 2019.
2. Elad Elrom, "The Blockchain Developer: A Practical Guide for Designing, Implementing, Publishing, Testing, and Securing Distributed Blockchain-based Projects", Apress, 2019.
3. Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller, and Steven Goldfeder, "Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction", Princeton University Press, 2016.

REFERENCE BOOKS

1. Andreas M. Antonopoulos, "Mastering Bitcoin: Programming the Open Blockchain", Shroff / O'Reilly, 2017.
2. Seema Shukla, Mehul Dhawan, Suyash Sharma, and Srinivas Venkatesan, "Blockchain Technology: Cryptocurrency and Applications", Oxford University Press.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PSO1	PSO2	PSO3
CO1	-	-	2	2	-	-
CO2	1	-	2	2	1	2
CO3	-	-	2	2	-	-
CO4	2	1	3	3	2	2
CO5	3	2	3	3	3	3
Score	6	3	12	12	6	7
COM	2	2	2	2	2	2

Course Code	MYPE510
Course Title	Ethical Hacking
L-T-P-C	3-0-0-3
Course Type	PE

COURSE OBJECTIVES

- To understand fundamental concepts and techniques of ethical hacking.
- To explore tools and methods used by attackers to exploit system vulnerabilities.
- To apply penetration testing phases: reconnaissance, scanning, exploitation, and post-exploitation.
- To develop secure systems by understanding and mitigating common attack vectors.
- To engage in hands-on ethical hacking labs simulating real-world attack scenarios.

COURSE CONTENT

Unit-I Introduction to Ethical Hacking
Ethical Hacking Fundamentals, Hacker Types, Hacking Phases, Cyber Laws And Ethical Issues, Footprinting Techniques, Reconnaissance Methods, Network Scanning (Nmap, Hping3), Enumeration, Vulnerability Assessment.

Unit-II System Hacking and Malware
Password Cracking, Privilege Escalation, Keylogging, Spyware And Rootkits, Trojans And Backdoors, Malware Analysis, OS Exploitation (Windows And Linux), Anti-Forensics, Clearing Tracks.

Unit-III Web And Network Attacks
Web Application Vulnerabilities (SQL Injection, XSS, CSRF), Session Hijacking, URL Manipulation, DNS Spoofing, ARP Poisoning, Man-In-The-Middle Attacks, Wireless Network Hacking, Wi-Fi Security Protocols.

Unit-IV Penetration Testing Framework
Penetration Testing Phases And Methodology, Metasploit Framework, Basic Exploit Development, Vulnerability Scanning Tools, Report Writing, Social Engineering Attacks.

Unit-V Defensive Techniques And Emerging Trends
Host Hardening, Patch Management, Firewalls, IDS/IPS, Honeypots, Cryptographic Techniques, Cloud Security Issues, IoT Threats, Ethical Concerns, Cybersecurity Case Studies.

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Demonstrate foundational knowledge of ethical hacking principles and practices	K3
CO2	Perform reconnaissance, scanning, and enumeration of target systems	K2
CO3	Exploit known vulnerabilities and demonstrate real-world attacks in a controlled environment	K4
CO4	Implement basic defensive strategies and analyze the effectiveness of countermeasures	K2
CO5	Apply penetration testing tools and methodologies in security assessments	K2

TEXT BOOKS

1. Daniel G. Graham, "Ethical Hacking: A Hands-on Introduction to Breaking In", 2nd Edition, No Starch Press, 2022.
2. Georgia Weidman, "Penetration Testing: A Hands-On Introduction to Hacking", No Starch Press, 2014.

REFERENCE BOOKS

1. Dafydd Stuttard and Marcus Pinto, "The Web Application Hacker's Handbook", Wiley, 2011.
2. William Stallings, "Network Security Essentials", 6th Edition, Pearson, 2023.
3. Patrick Engebretson, "The Basics of Hacking and Penetration Testing", 2nd Edition, Syngress, 2013.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PSO1	PSO2	PSO3
CO1	3	3	3	3	-	-
CO2	2	3	3	3	2	-
CO3	3	2	-	-	2	-
CO4	-	2	-	-	2	-
CO5	3	2	3	3	2	-
Score	11	12	9	9	8	-
COM	3	2	3	3	2	-

Course Code	MYPE511
Course Title	Digital Forensics
L-T-P-C	3-0-0-3
Course Type	PE

COURSE OBJECTIVES

- To understand the concept of digital forensics and its terminology.
- To understand technical and legal aspects of computer crime investigations.
- To have in-depth understanding of attacks and their counter measures.
- To know how to apply defense through data analysis and file system analysis.
- To learn the procedures of recovering computer evidence and the seize process.

COURSE CONTENT

Unit-I Introduction
Computer forensics fundamentals, Benefits of forensics, computer crimes, computer forensics evidence and courts, legal concerns and private issues.

Unit-II Computing Fundamentals
Understanding Computing Investigations, Procedure for corporate High-Tech investigations, understanding data recovery work station and software, conducting investigations.

Unit-III File Systems
File Systems, Various Types of File Systems, Introduction to Storage Layers, Hard Disk Drive, Forensic Analysis of File Systems. Preparation Steps for Forensic Analysis.

Unit-IV Mobile Forensics
Processing crimes and incident scenes, securing a computer incident or crime, seizing digital evidence at scene, storing digital evidence, obtaining digital hash, reviewing case. Mobile Phone Forensics and Call Forging, Forensic Procedures, Android Forensics, Mobile Forensic Tools, Computer Forensics Software Tools, Computer Forensics Hardware Tools, Various Computer/Digital Forensic Tools.

Unit-V Intrusion Detection System
Introduction to Intrusion Detection System, Types of Intrusion Detection System, Advantages and Disadvantages of IDS, Understanding Network Intrusions and Attacks, Recognizing Pre-Intrusion/Attack Activities, Port Scans, Address Spoofing, Attacking with Trojans, Viruses, and Worms, Understanding Password Cracking, Understanding Technical Exploits, Collecting Network-Based Evidence, Investigating Routers, Network Protocols.

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Understand requirements of digital forensics its related terminology	K2
CO2	Explain technical and legal aspects of computer crime investigations	K2
CO3	Know the limitations of law and its enforcement agencies	K1
CO4	Illustrate forensic analysis and data analysis on file systems	K2
CO5	Apply techniques for finding, preserving, presenting, and extracting information from the digital devices	K3

TEXT BOOKS

1. Joakim Kavrestad, Marcus Birath, and Nathan Clarke, “Fundamentals of Digital Forensics: A Guide to Theory, Research and Applications”, 3rd Edition, Springer, 2024.
2. Amah Nnachi Lofty, “Digital Forensics: Investigating Cyber Crimes”, Redshine Publication, 2024.

REFERENCE BOOKS

1. Akashdeep Bhardwaj and Keshav Kaushik, “Practical Digital Forensics”, BPB Publications, 2023.
2. Manas Kamal Bhuyan, “Computer Vision and Image Processing: Fundamentals and Applications”, 2019.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PSO1	PSO2	PSO3
CO1	3	-	-	-	-	-
CO2	3	-	-	-	-	-
CO3	3	-	-	-	-	-
CO4	3	2	2	-	-	2
CO5	3	3	2	-	-	2
Score	15	5	4	-	-	4
COM	3	3	2	-	-	2

Course Code	MYPE512
Course Title	Responsible AI
L-T-P-C	3-0-0-3
Course Type	PE

COURSE OBJECTIVES

- To understand the evolution of AI capabilities and identify ethical and societal implications.
- To analyze bias, fairness, and transparency in AI systems.
- To apply interpretability and explainability techniques to AI models.
- To explore privacy-preserving mechanisms and adversarial robustness.
- To understand global regulations and implement responsible AI practices in domains like healthcare and education.

COURSE CONTENT

Unit-I Foundations of Responsible AI and Ethical Principles
AI capabilities improvement, Introduction to Responsible AI, Need for ethics in AI: AI for society and humanity, Risks from AI models: Bias, toxicity, adversarial attacks, misuse, rogue AGI, Principles of Responsible AI: Fairness, Accountability, Transparency, Privacy, Safety, Human-centricity, Interpretability.

Unit-II Fairness and Bias in AI
Sources and types of biases in data and models, Dataset limitations and EDA, Fairness definitions: Group fairness, Individual fairness, Counterfactual fairness, Bias mitigation techniques: Preprocessing, Inprocessing, Postprocessing, Tools to evaluate and measure fairness.

Unit-III Interpretability and Explainability
Need for Explainable AI, Intrinsic interpretable models (e.g., linear models, decision trees), Post-hoc techniques: LIME, SHAP, GradCAM, Explainability through simplification, visualization, and causality, Mechanistic interpretability, model editing, and probing, Critiques of transparency for AI safety.

Unit-IV Privacy, Adversarial Robustness, and AI Safety
Adversarial attacks in Vision, NLP (e.g., AlphaGo agents), Model poisoning attacks: Trojans and backdoors, ML lifecycle implications for safety, Differential privacy and Federated learning, Causal modeling: Graphical models, counterfactuals, interventions, Distribution shift and generalization, AI alignment and instrumental convergence risks.

Unit-V Regulation, Applications, and Domain Case Studies
Regulation landscape: DPDP Act (India), GDPR (EU), EU AI Act, US Presidential Orders, Ethical reviews, informed consent, participatory design, RAI in domains: Recommendation systems, Healthcare, Education, Hiring, Computer Vision, NLP, AGI: Definitions, risks, future timelines, Paper readings, panel discussions, fireside chats (optional integration).

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Understand and articulate ethical and responsible principles in AI systems	K3
CO2	Identify and mitigate algorithmic biases in datasets and models	K3
CO3	Apply interpretability techniques like SHAP, LIME, and GradCAM	K4
CO4	Evaluate and implement privacy-preserving AI models	K4
CO5	Analyze AI policy frameworks and domain-specific RAI applications	K4

TEXT BOOKS

1. Mark Coeckelbergh, "AI Ethics", MIT Press, 2020.
2. Christoph Molnar, "Interpretable Machine Learning", Leanpub, 2022.

REFERENCE BOOKS

1. Michael Kearns and Aaron Roth, "The Ethical Algorithm", Oxford University Press, 2019.
2. Stuart Russell, "Human Compatible: Artificial Intelligence and the Problem of Control", Viking, 2019.
3. Solon Barocas, Moritz Hardt, and Arvind Narayanan, "Fairness and Machine Learning", fairmlbook.org, 2023.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PSO1	PSO2	PSO3
CO1	3	3	3	3	2	2
CO2	2	3	3	3	3	2
CO3	3	2	3	3	3	2
CO4	2	2	2	3	3	3
CO5	3	3	3	3	3	2
Score	13	13	14	15	14	11
COM	3	3	3	3	3	2

Course Code	MIVA502
Course Title	Technical Writing and Communication Skills
L-T-P-C	1-0-2-0
Course Type	VA

COURSE OBJECTIVES

- To understand the fundamentals and scope of technical and scientific writing.
- To understand the structure and components of various scientific documents.
- To understand the role of grammar, punctuation, and vocabulary.
- To understand the principles of effective presentation design and delivery.
- To understand the dynamics of professional communication.

COURSE CONTENT

Unit-I: Foundation of Technical and Scientific Writing
Introduction to Technical Writing: Nature and Scope
Various Forms of Scientific Writing: Theses and Dissertations, Technical Papers, Manuals and Project Reports Review Articles, Components of Thesis and Research Communication, Review of Literature, Results, Discussion, Conclusion, Common Abbreviations in Technical Writing, Email and Technical Correspondence

Unit-II: Writing Strategies and Research Communication
Writing Abstracts, Summaries, Précis, and Synopsis
Structure and Features of Technical Reports, Types and Purpose of Reports, Data Collection and Analysis Graphic Aids (Charts, Tables, Diagrams), Drafting and Revising Scientific Texts
Writing Research Papers: Basic Guidelines, Style and Tone in Research Writing
Referencing and Documentation Styles, Editing and Proofreading Techniques

Unit-III: Grammar in Professional Writing
Grammar in Professional Writing: Tense, Punctuation and Sentence Structure, Common Errors in Scientific Writing, Technical and Academic Word Usage. AI tools for writing and paraphrasing (such as ChatGPT, QuillBot, etc.).

Unit-IV: Presentation Skills
Planning and Structuring Presentations, Slide Design and Use of Visual Aids, Voice Modulation and Body Language, Handling Q & A and Stage Fear, Time Management During Presentations

Unit-V: Professional and Interpersonal Communication
Group Discussion Practice and Feedback, Technical Talk, Panel Discussion, "Teach a Concept" Interview Skills

LAB CONTENT

The experiments may be performed as per the course contents mentioned above.

COURSE OUTCOMES

CO	Description	Level
CO1	To develop the ability to produce clear, structured, and professional technical documents.	K3
CO2	To develop proficiency in summarizing complex information.	K2
CO3	To develop competence in using correct grammar, punctuation, and domain-specific vocabulary.	K4
CO4	To develop effective oral presentation skills.	K2
CO5	To develop professional interpersonal communication skills.	K2

TEXT BOOKS

1. Meenakshi Raman and Sangeeta Sharma, *Technical Communication: Principles and Practice*, 3rd Edition, Oxford University Press, 2015.
2. Philip Rubens (Ed.), *Science and Technical Writing: A Manual of Style*, 2nd Edition, Routledge, 2001.

REFERENCE BOOKS

1. Heather Silyn-Roberts, *Writing for Science and Engineering: Papers, Presentations and Reports*, 2nd Edition, Butterworth-Heinemann (Elsevier), 2012.
2. Adrian Wallwork, *English for Writing Research Papers*, 2nd Edition, Springer, 2016.

CO to PO/PSO Mapping

	PO1	PO2	PO3	PSO1	PSO2	PSO3
CO1	3	2	2	3	-	-
CO2	2	3	2	2	3	-
CO3	3	2	3	3	2	1
CO4	2	3	2	2	2	2
CO5	3	2	3	3	2	2
Score	13	12	12	13	9	5
COM	3	2	2	3	2	2

Course Code	MCPR501
Course Title	Minor Project
L-T-P-C	0-0-4-2
Course Type	PR

COURSE OBJECTIVES

- To explore project domain and state objectives clearly and concisely.
- To learn the state-of-art methods for the project domain.
- To survey research problems and derive methodologies to solve the problem.
- To analyze state-of-the-art techniques to solve real-world problems.
- To demonstrate and implement the proposed solution.

COURSE CONTENT

The minor project work is designed for one semester duration as a single project involving detailed literature survey, experimentation plan and its implementation.

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Demonstrate a sound technical knowledge of the selected project topic.	K2
CO2	Survey research studies, find research gaps, and formulate a complex engineering problem.	K2
CO3	Experiment with state-of-the-art methods and identify the available solutions.	K4
CO4	Analyze and compare the available solutions.	K4
CO5	Plan, propose, and implement the proposed solution.	K6

CO to PO/PSO Mapping

	PO1	PO2	PO3	PSO1	PSO2	PSO3
CO1	3	3	3	3	2	3
CO2	3	3	3	3	2	3
CO3	3	3	3	3	2	3
CO4	3	3	3	3	2	3
CO5	3	3	3	3	3	3
Score	15	15	15	15	11	15
COM	3	3	3	3	2	3

SEMESTER III

Course Code	MCPR601
Course Title	Dissertation I
L-T-P-C	0-0-0-9
Course Type	PR

COURSE OBJECTIVES

- To identify the complex research/engineering problems.
- To understand different methodologies for documentation as report.
- To demonstrate systematic understanding and sound technical knowledge of their topic of research.
- To examine the key stages to analyze a research solution.
- To understand the process of carrying out research and report the result with reference to existing literature.

COURSE CONTENT

The Dissertation I work is designed for one semester duration as a literature survey/review on the selected research problem from the area of interest which shall benefit the society.

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Select a research/engineering problem of their interest for societal benefit.	K2
CO2	Understand existing solutions, if any, and document informally (scientific notes) and formally (report).	K2
CO3	Have an in-depth technical knowledge on their topic of research.	K1
CO4	Analyze research solutions on different parameters.	K4
CO5	Understand the process of methodical research to deliver the result or solve the problem.	K2

CO to PO/PSO Mapping

	PO1	PO2	PO3	PSO1	PSO2	PSO3
CO1	3	2	2	2	2	3
CO2	3	3	2	2	2	3
CO3	3	2	3	3	2	3
CO4	3	3	3	3	3	3
CO5	3	3	1	1	1	3
Score	15	13	11	11	10	15
COM	3	3	2	2	2	3

SEMESTER IV

Course Code	MCPR601
Course Title	Dissertation II
L-T-P-C	0-0-0-12
Course Type	PR

COURSE OBJECTIVES

- To evaluate the methods and knowledge to solve the specific research/engineering problem.
- To select the suitable method which leads to appropriate results.
- To design research/engineering solutions to the selected problem for the benefit of society.
- To formally document the designed research/engineering solution.
- To develop strong communication skills to justify the research/engineering work in front of technically qualified audience.

COURSE CONTENT

The Dissertation II work is designed for one semester duration to design/develop a research solution for a selected research problem of choice to benefit society.

COURSE OUTCOMES

After the completion of this course, students will be able to:

CO	Description	Level
CO1	Evaluate different methods to solve the specific research/engineering problem.	K5
CO2	Decide the suitable method comparatively that leads to appropriate results.	K3
CO3	Design research/engineering solution to benefit society.	K6
CO4	Report/Document their designed solution.	K2
CO5	Justify their research/engineering solution in front of technically qualified audience.	K2

CO to PO/PSO Mapping

	PO1	PO2	PO3	PSO1	PSO2	PSO3
CO1	3	3	3	3	3	3
CO2	3	3	3	3	3	3
CO3	3	3	3	3	3	3
CO4	3	3	3	3	3	3
CO5	3	3	3	3	3	3
Score	15	15	15	15	15	15
COM	3	3	3	3	3	3